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1. Creating and Managing Custom Activities

1.1. Introduction to Custom Activity

Custom activities are activities that can be designed as per your requirements. The activities created can be utilized in the process flows for an organization.

There are two types of custom activities:

a) custom activity of type custom activity,

b) custom activity of type subflow.

A custom activity of type custom activity is designed by uploading JSON files, Javascript, and JAR files that contain the configuration details required for that custom activity.

A custom activity of type subflow allows you to design a process flow as an activity.

1.2. Creating New Custom Activity Configuration

Follow the below steps for creating a new Custom Activity configuration.

1. Click the Burger menu and navigate to **Manage** > **Configuration Management**.
1. Click **Custom Activity** in the configuration entity panel.

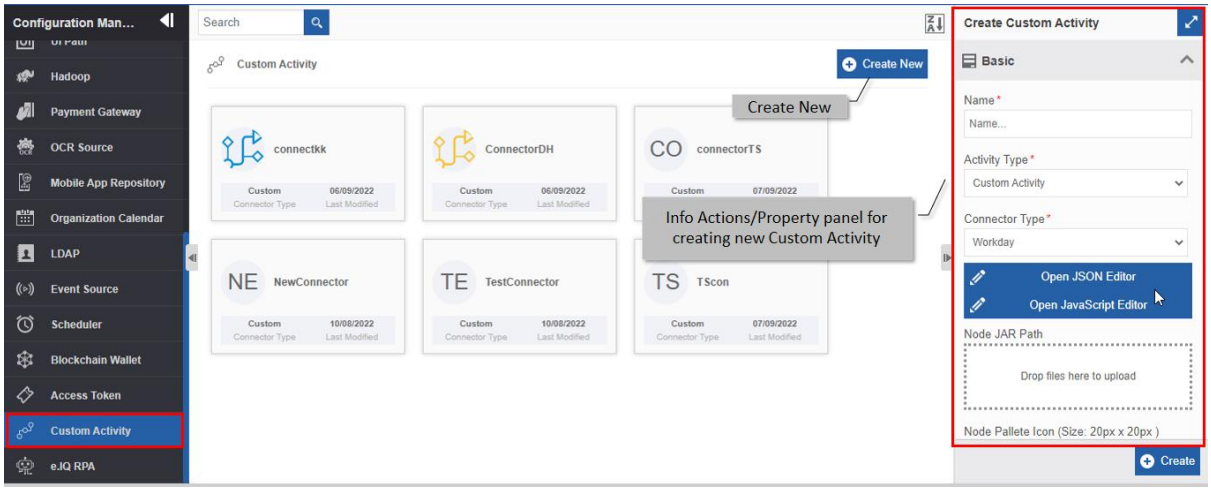
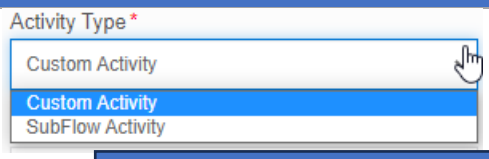
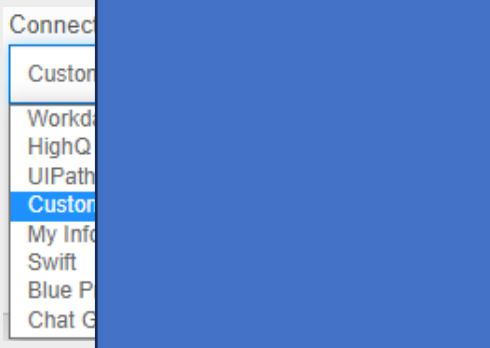


Figure 1.1: Creating new Custom Activity

2. Click ⊕**Create New.**
3. Enter the configuration details in the **Create Custom Activity** panel as explained below.

Field	Description
Name*	Enter the name of the custom activity. Character limit: 50. Data type: Alphanumeric and underscore.
Activity Type*	Select the activity type.

Field	Description
	 **Custom** as per your configuration details. **Sub Flo** can be used as an activity. Refer to
Connector Type*	Select t  Refer to more details on connectors.
Open JSON Editor	Click to he JSON code. The info ucture of the activity is coded in the J The noc or will appear on the Custom Activity Refer to for more information.
Open JavaScript Editor	Click to aste the JavaScript code. The det e input content are coded in the java Refer to tails for more information.
Node JAR Path	Upload The pro the time of execution of the custom Refer to ails for more information.
Node Pallete Icon*	Upload listing purposes. This icon appears Only SV used to define vector-based graphics The ma loaded is 20px X 20 px
Node Icon*	Upload activity in the designer area when t o the process flow designer area.

Field	Description
	Only SVG file format is supported. SVG is used to define vector-based graphics for the Web. The maximum size of the image that can be uploaded is 40px X 40px
Node Accordion Icon	Upload node accordion icon. This icon appears on the Info Actions panel for the activity configuration category accordion. Only SVG format is supported. SVG is used to define vector-based graphics for the Web. The maximum size of the image that can be uploaded is 20px X 20 px.
Description	Write a brief description of the custom activity. Character limit: 1000 characters Data type: Alphanumeric and symbols.

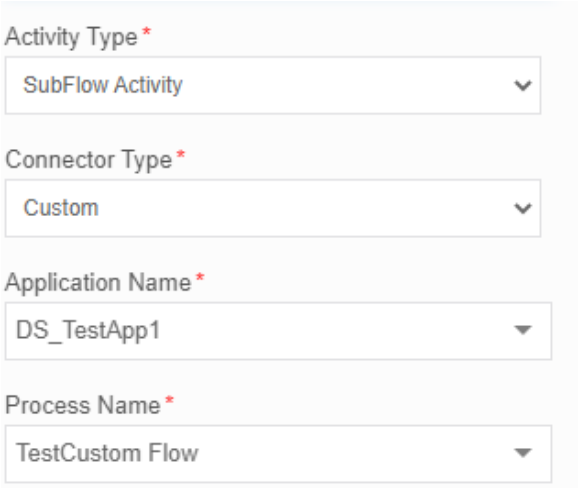
☞ ** Mandatory (required) fields.*

4. Click **Create** on the bottom right of the page and the custom activity gets created with the details entered.

1.2.1. Creating Subflow for Custom Activity

To create a process flow as a custom activity, follow the below steps.

1. Navigate to **App Studio** > **Process Flow** module.
2. Create and configure the process flow details and entities as per your needs.
3. Deploy the process flow.
4. Navigate to **Manage** > **Configuration Management** > **Custom Activity**.
5. Click *Create New* and create a Custom Activity by selecting **Activity Type** as **Sub Flow Activity**. Application Name drop-down is displayed. The application name is displayed only if one of the process flow within that application is deployed.
6. Select the application name. The Process flow list is displayed. Only the deployed process flows are listed in the drop-down.
7. Select the process flow that you want to define as the custom activity.



8. For other configurations, go to step 3 details and configure accordingly, and finally click Create for creating the subflow type custom activity.

1.2.2. Custom Activity Connector Types

Connectors are used to achieve specific requirements in your activity. Some connectors are part of the Platform, however, you can include external connectors for utilizing the features of the same.

The connector types supported in the custom activities are explained below.

Blue Prism: Selecting Blue Prism allows you to create a custom activity that interacts with the Blue Prism application via APIs. When you create a Blue Prism custom activity, it will be listed in the Blue Prism category of the Activities.

Chat GPT: Selecting Chat GPT allows you to create a custom activity that interacts with the Chat GPT application via APIs. When you create a Chat GPT custom activity, it will be listed in the Chat GPT category of the Activities.

Custom: Select the option "Custom" if you are defining a custom activity or if you are creating a subflow custom activity within the Platform.

HighQ: Selecting HighQ allows you to create a custom activity that interacts with the HighQ application via APIs. When you create a HighQ custom activity, it will be listed in the HighQ category of the Activities.

My Info: Selecting My Info allows you to create a custom activity that interacts with the My Info application via APIs. When you create a My Info custom activity, it will be listed in the My Info category of the Activities.

Swift: Selecting Swift allows you to create a custom activity that interacts with the Swift application via APIs. When you create a Swift custom activity, it will be listed in the Swift category of the Activities.

UIPath: Selecting UIPath allows you to create a custom activity that interacts with the UIPath application via APIs. When you create a UIPath custom activity, it will be listed in the UIPath category of the Activities.

Workday: Selecting Workday allows you to create a custom activity that interacts with the Workday application via APIs. When you create a Workday custom activity, it will be listed in the Workday category of the Activities.

1.3. Viewing and Editing Custom Activity

1. Click the Burger menu and navigate to **Management** > **Configuration Management** > **Custom Activity**.
2. Click the Custom Activity card to view the details of the selected Custom Activity. The details of the Custom Activity appear in the Info Actions panel (**Edit Custom Activity**).

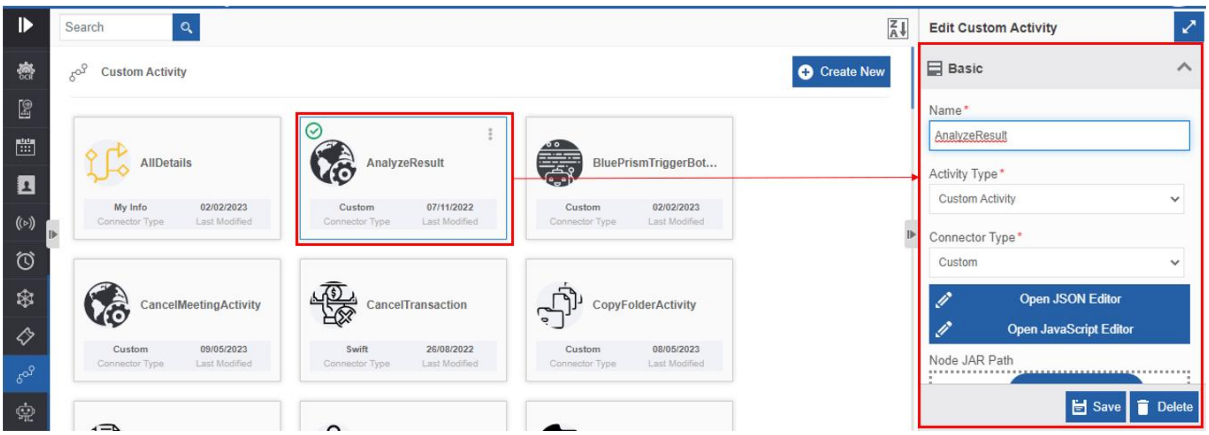


Figure 1.2: Editing Custom Activity details

3. Edit the Custom Activity details as needed.
4. Click **Save**.

1.4. Duplicating Custom Activity

Follow the below steps for duplicating an existing custom activity.

1. Click the Burger menu and navigate to **Manage** > **Configuration Management**.
2. Click **Custom Activity**. The list of all custom activities is displayed.
3. Hover over any custom activity card. Three dots appear on the upper right side of the card.
4. Click the three dots. More Actions appear.

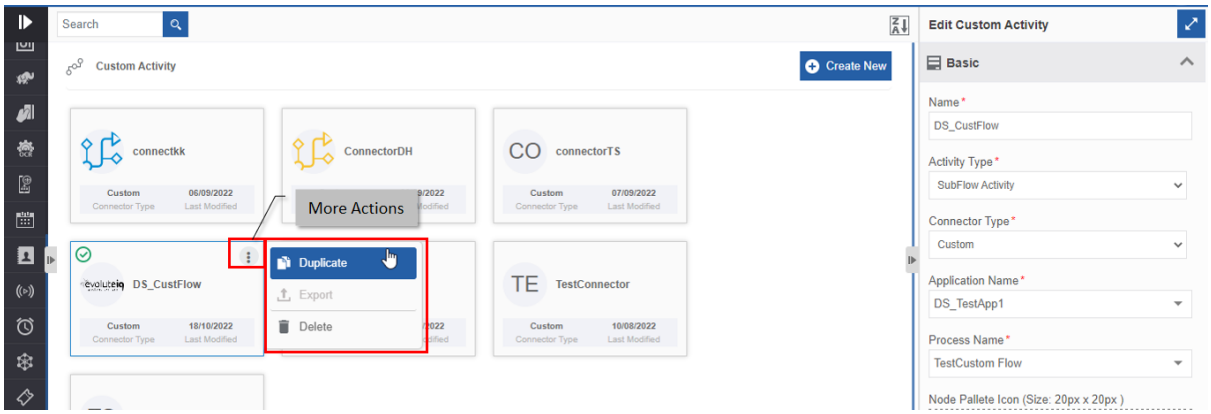


Figure 1.3: Duplicating the custom activity

5. Click **Duplicate**. A confirmation pop-up appears.

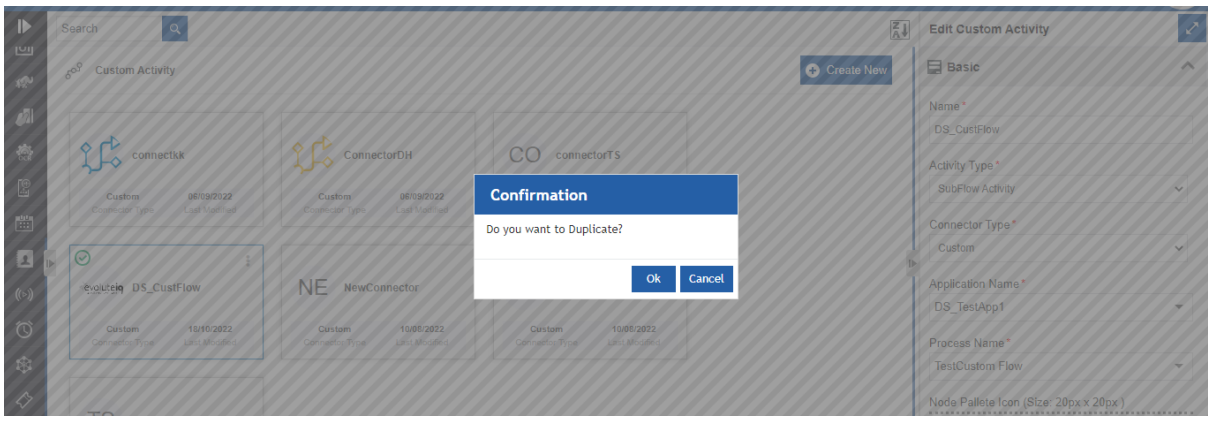


Figure 1.4: Duplicate confirmation

6. Click **Ok** for duplicating the custom activity (or you can click **Cancel** to cancel the duplicate action). A **Success** message appears on the successful duplication of the custom activity.

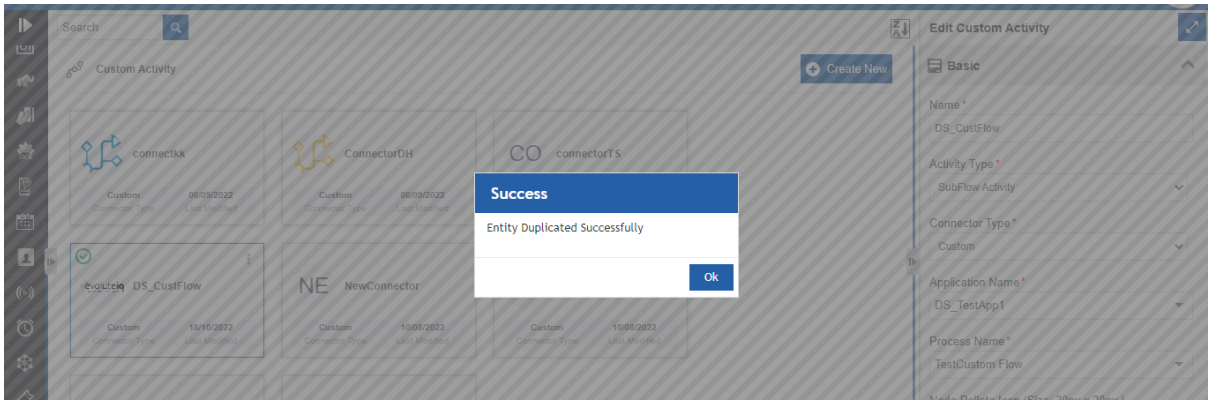


Figure 1.5: Duplicate success message

7. Click **Ok**. A duplicate copy of the custom activity appears on the custom activity page with the same custom activity name suffixed with "**_copied**".

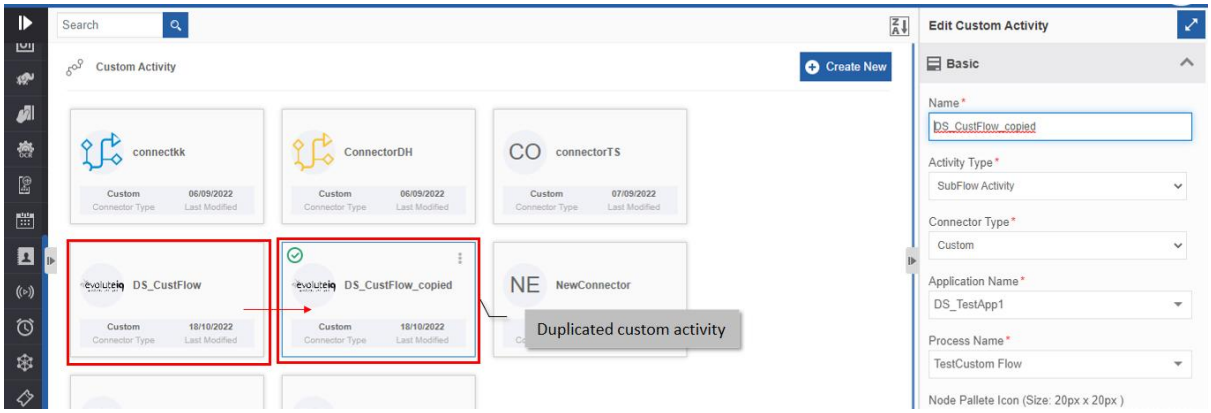


Figure 1.6: The duplicated custom activity

1.5. Exporting a Custom Activity

You can export a custom activity as a package to your local system.

1. Click the Burger menu and navigate to **Management** > **Configuration Management** > **Custom Activity**.
2. Hover over a custom activity name card and click the three dots (More Actions).

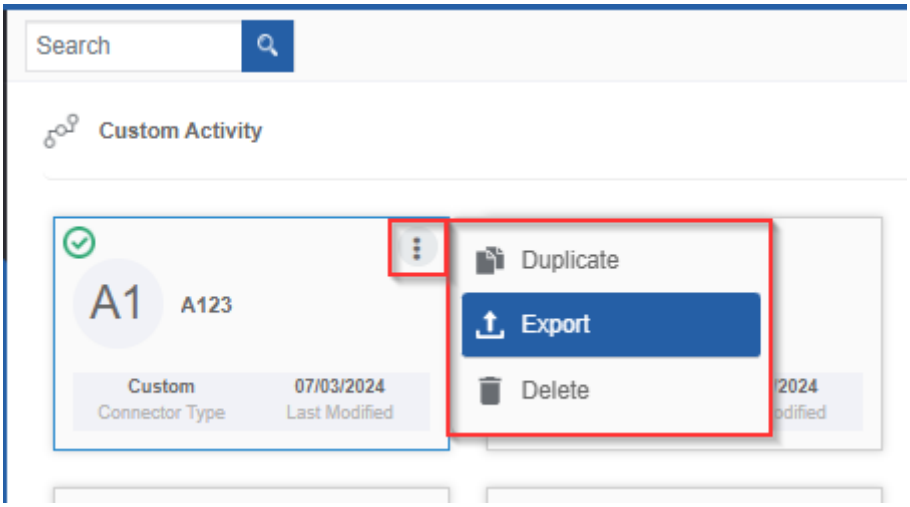


Figure 1.7: Exporting Custom Activity

3. Click **Export**. The selected custom activity gets exported to your local system in the .pkg format.

1.6. Importing Web App

You can import a custom activity package into the Custom Activity section for utilization.

1. Click the Burger menu and navigate to **Management** > **Configuration Management** > **Custom Activity**. The list of all custom activities in the platform appears.

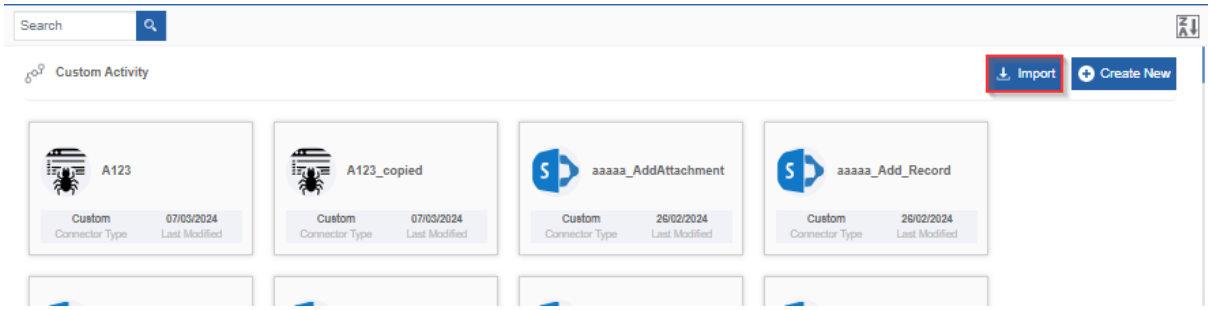


Figure 1.8: Import custom activity

2. On the upper right click **Import**.

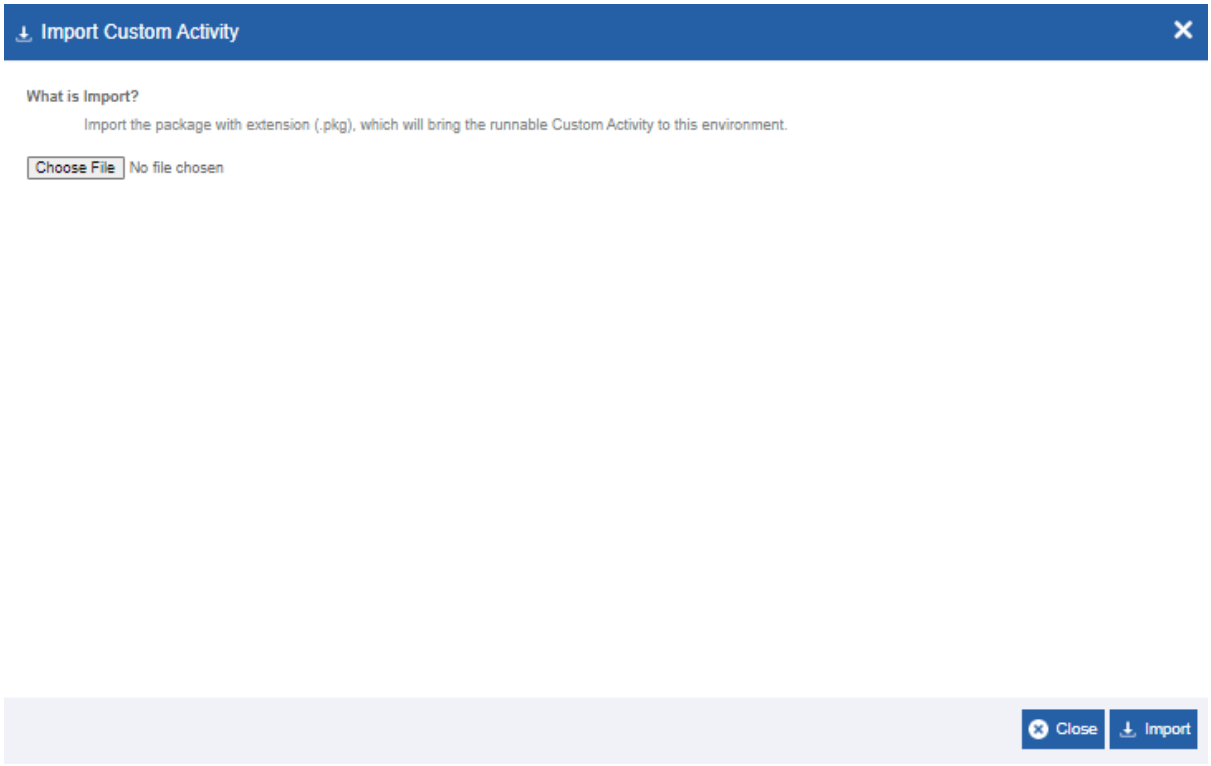


Figure 1.9: Importing Custom Activity - file selection

3. Click **Choose File** and select the file (in .pkg format) that you want to import. If the import package name and matches with any of the existing custom activity, import filed message appears with an option to rename the package name or to overwrite the existing custom activity in the platform.

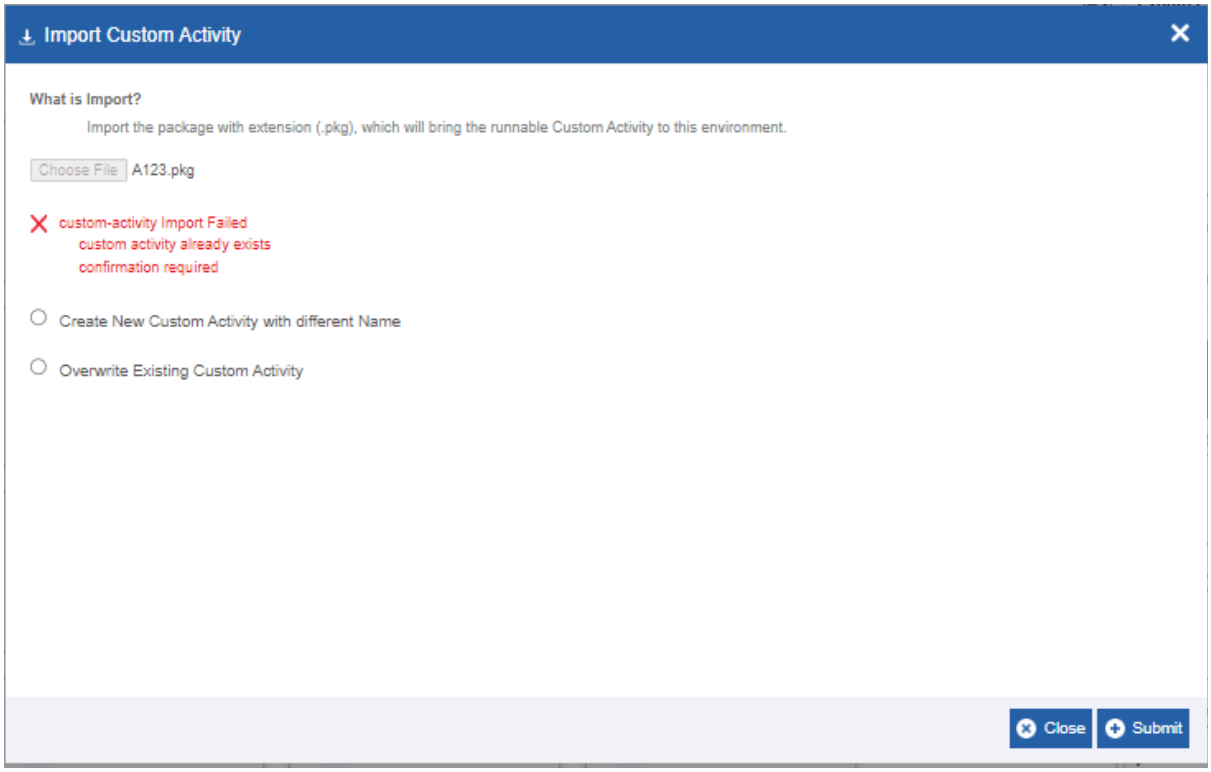
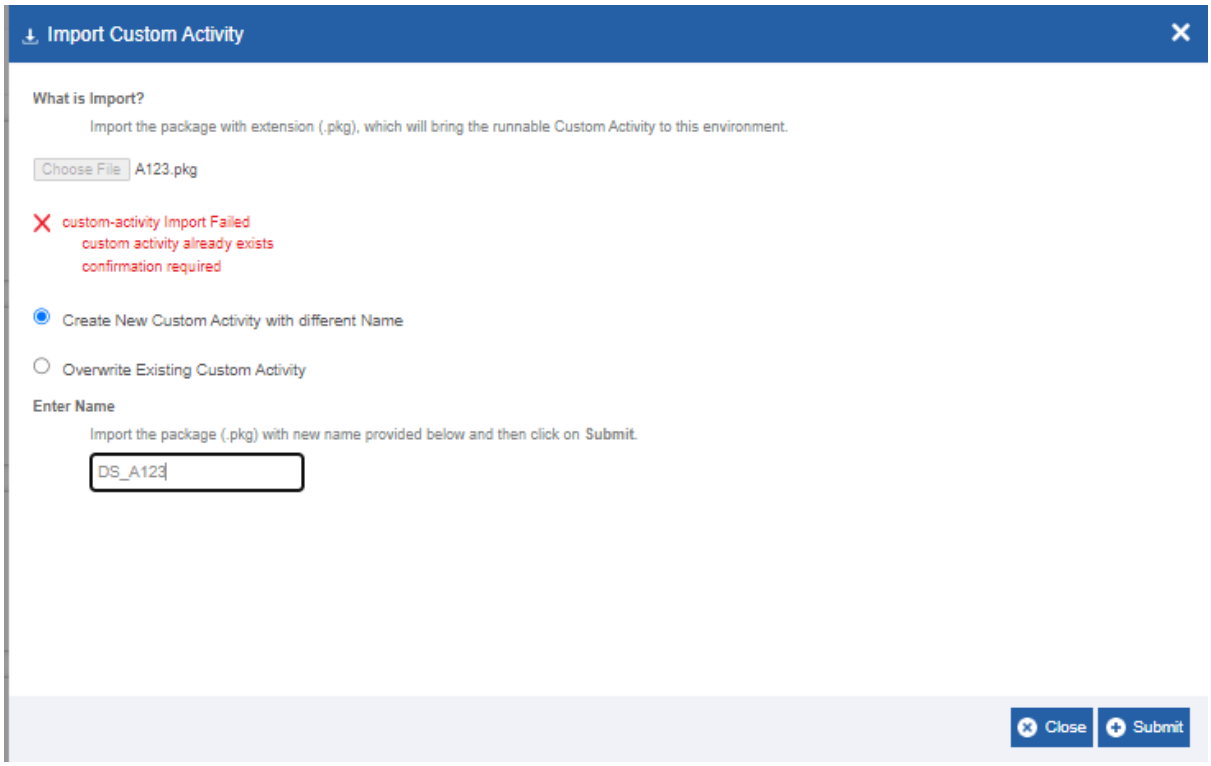


Figure 1.10: Import custom activity import failed message with options

- **Create new Custom Activity with different name**: Click this option if you want to import the same package with a different name.



Enter the new name in the **Enter Name** field.

Overwrite Existing Custom Activity: Click this option to overwrite the existing custom activity with the import file with the same name as in the package.

4. Click **Import**. A Success message appears on successful import of the package.

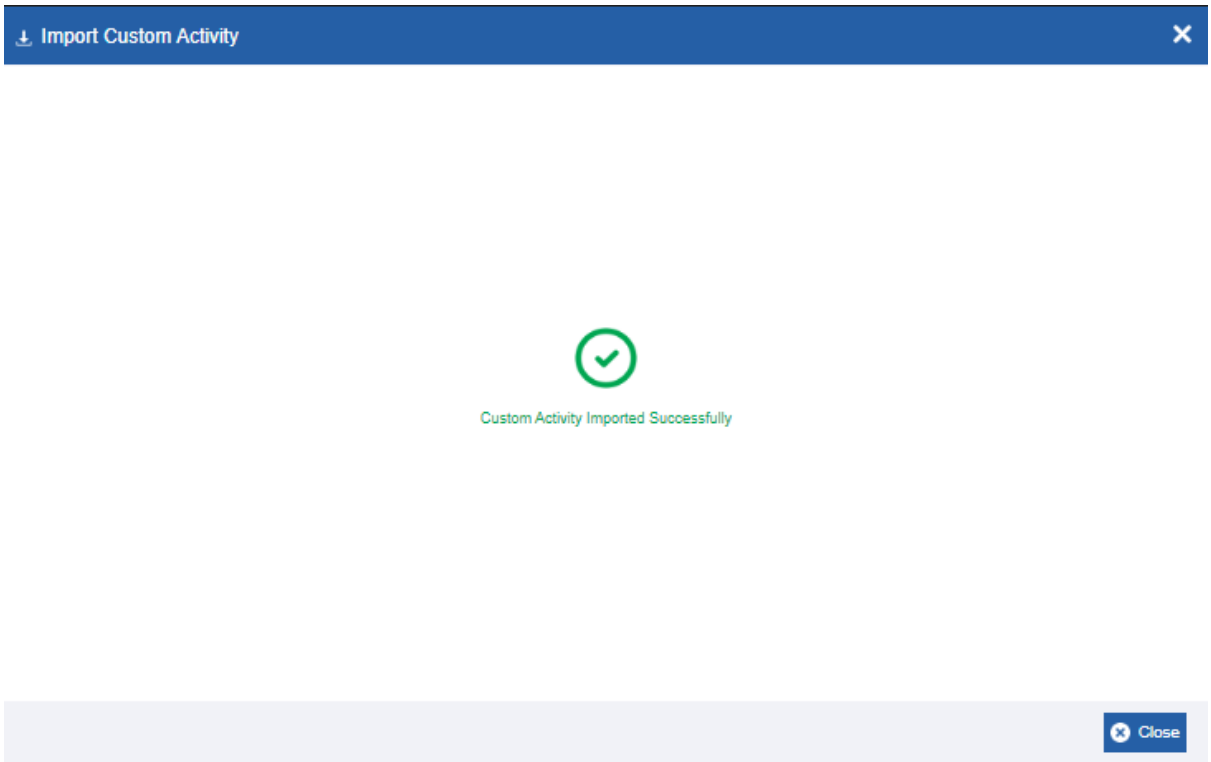


Figure 1.11: Import success message

5. Click **Close**. The package is imported to the platform.

1.7. Deleting Custom Activity

1. Click the Burger menu and navigate to **Manage > Configuration Management**.
2. Click **Custom Activity**. The list of all custom activities is displayed.
3. Click the custom activity name card that is to be deleted. The lower-right of the page displays the **Delete** button.

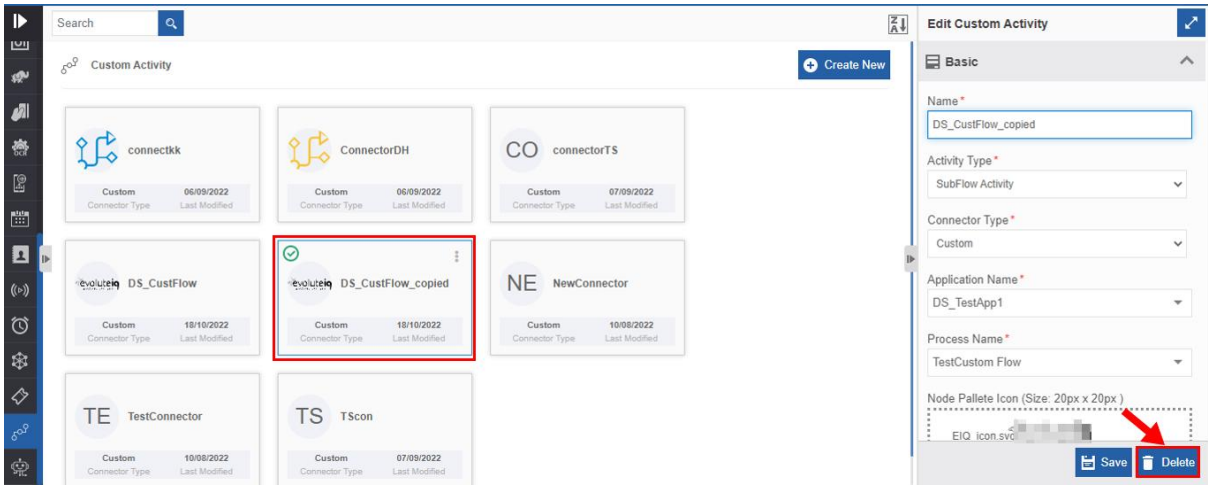


Figure 1.12: Deleting custom activity

4. Click **Delete**. A **Confirmation** pop-up for delete appears.

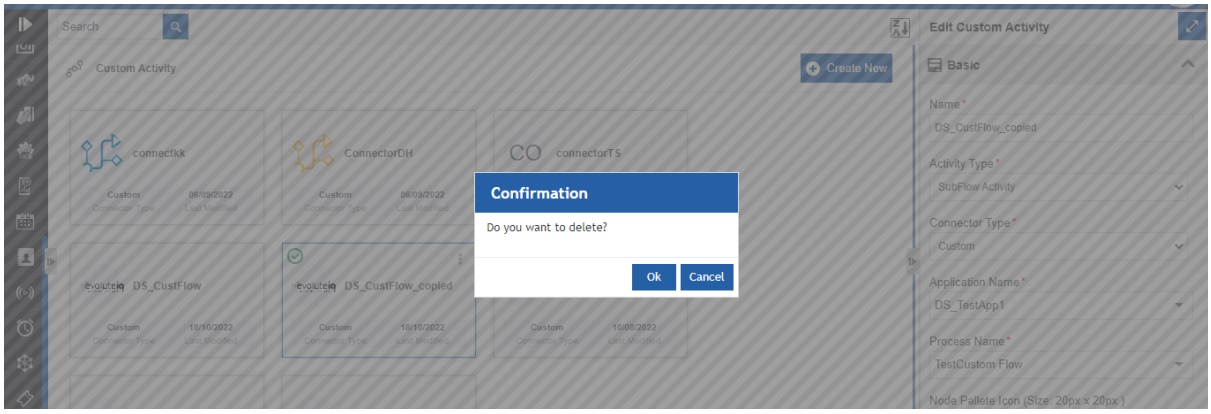


Figure 1.13: Custom Activity Delete confirmation

5. Click **Ok** for deleting the custom activity.

   Or
   Click **Cancel** to cancel the action.

Alternatively, you can follow the below steps to delete the custom activity:

i. Click the Burger menu and navigate to **Manage** > **Configuration Management** > **Custom Activity**

ii. Hover over the custom activity card. Three dots appear on the upper right side of the card.

iii. Click the three dots. More Actions appear.

iv. Click **Delete** and follow step 5 in the above procedure.

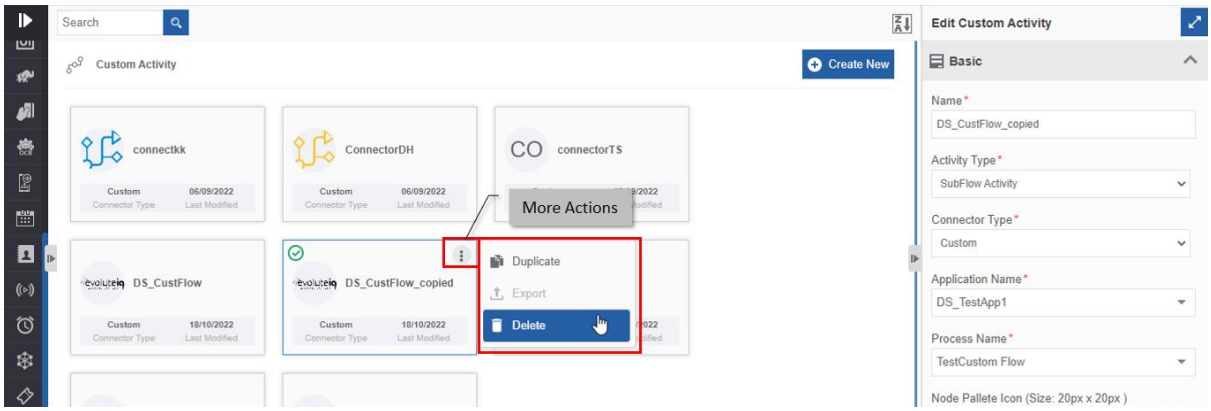


Figure 1.14: Delete action in More Actions

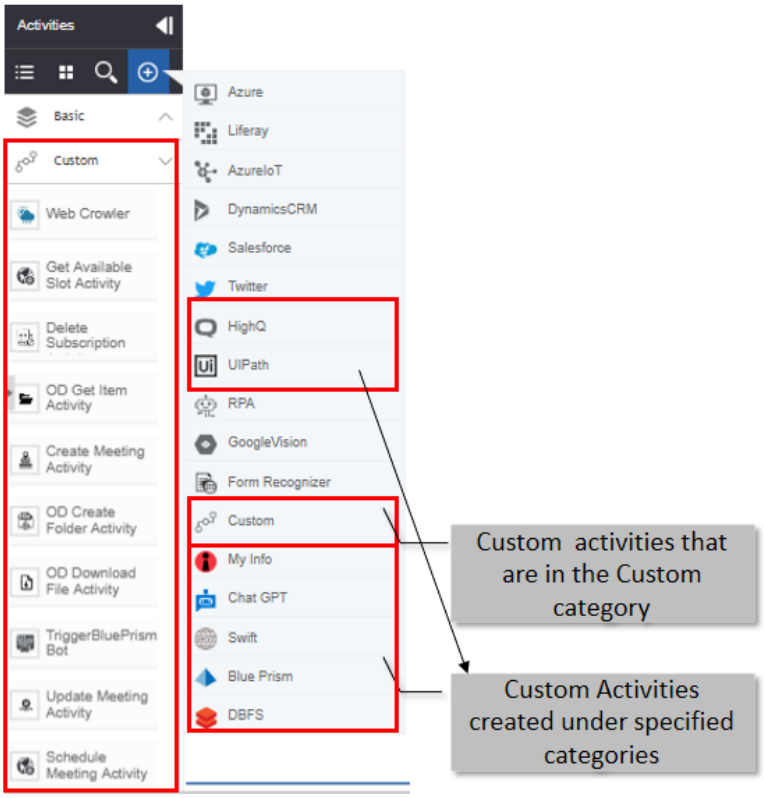
2. Custom Activities in Process Flows

2.1. Adding and Viewing Custom Activities

The custom activities created in the Manage > Configuration Management > Custom Activities can be accessed through the process flow Activities panel.

1. Navigate through **App Studio** > **Process Flows**.
2. In the Process Design > **Activities** panel click "+" for viewing the advanced connectors.
3. Click **Custom**. All the custom-defined activities get added to the Activities panel under the **Custom** accordion.

   The custom activities that are of type "Custom" are displayed below the Custom category. Other custom activities are listed below the selected custom category type.
   Custom activities that are in different categories will be listed under that category along with the default activities (if any) in that category.



2.2. Custom Activity in Process Flow

To use the custom activity, you need to add the custom activity category to the activities list and then drag the activity to the process designer.

Based on the type of custom activity added (Sub Flow Activity Type or Custom Activity Type) to the process flow, the configuration details are different.

The sections included in Info Actions for the custom activity of type Subflow are Basic, Trigger Subflow, and Error Handling. Refer to Configuring Subflow Activity.

The sections included in Info Actions for the Custom activity of type custom activity are Basic and the configurations are coded as in the JSON file. Refer to Configuring Custom Activity.

2.2.1. Configuring Subflow Activity

1. Add the Custom Activity category to the Activity list. Refer to Adding and Viewing Custom Activities.
2. Design the process flow by dragging the subflow custom activity as needed.

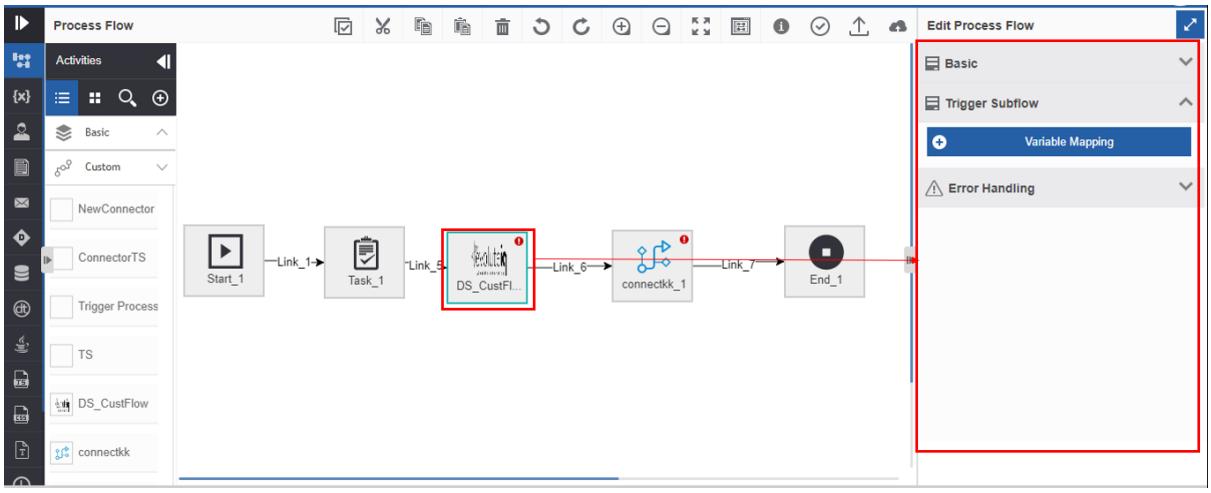
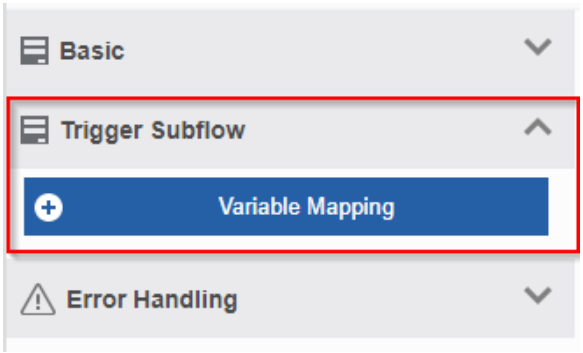


Figure 2.1: Custom subflow activity in process flow and the configurations

3. Click the activity. Configuration details are displayed on the Info Actions panel.
4. Click **Basic** accordion.
5. Provide **Name** and **Description** for the activity.

6. Click **Trigger Subflow** accordion.



   i. Click ⊕ **Variable Mapping**. Variable mapping pop-up appears.

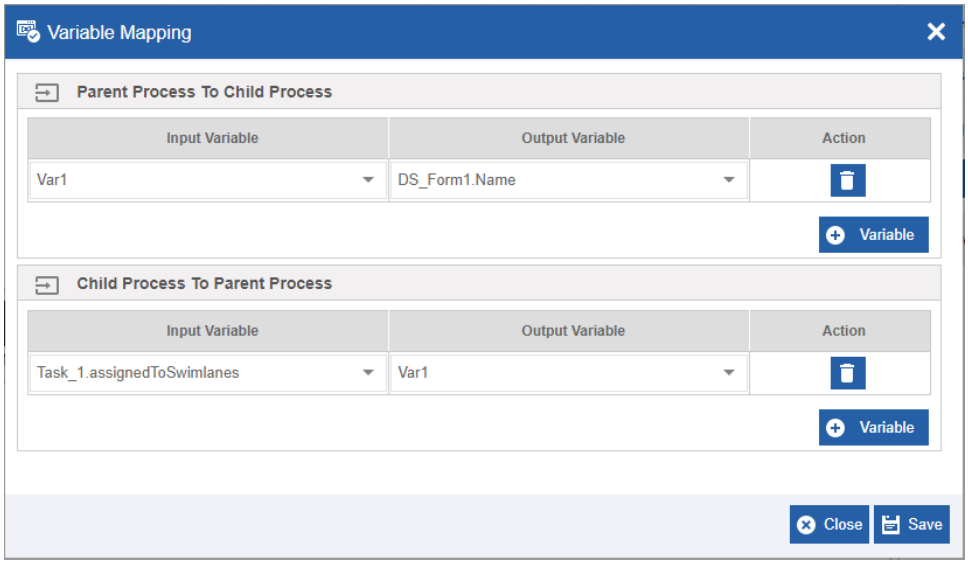


Figure 2.2: Variable Mapping

ii. In the Parent Process To Child Process, click **Variable** and select **Input Variable** (parent process variable) and **Output Variable** (child process variable). This is done to pass the data from the parent process to the child process, that is, custom subflow activity.

iii. In the Child Process To Parent Process, click **Variable** and select **Input Variable** (child process) and **Output Variable** (parent process). This is done to pass the data from the child process, that is custom subflow activity to the parent process.

iv. Click ⊕ **Variable**, to add more variables in each section.

v. Click the delete icon to remove an input or output variable mapping.

vi. Click **Save**.

7. Click **Error Handling** accordion.

Figure 2.3: Error Handling

i. Check the **Continue If Error** check box if you want to continue the workflow even if the error occurs.
Else uncheck the checkbox to break or stop the workflow execution when an error occurs.

ii. Select a variable from the **Error Variable** drop-down. The variable should be of data type, alphanumeric. Error variable stores the first line of the error.

8. Click **Create/Save** to save the changes.

2.2.2. Configuring Custom Activity

Custom activity configurations depend on the JSON file, JavaScript file, and the JAR file that you uploaded while creating the custom activity in the **Manage** > **Configuration Management** > **Custom Activity**.

1. Add the Custom Activity category to the Activity list. Refer to Adding and Viewing Custom Activities.
2. Design the process flow by dragging the custom activity as needed. The custom activities appear below the Custom category in the Activities pane. The name of the custom activity is the node name that you have coded in the JSON file.

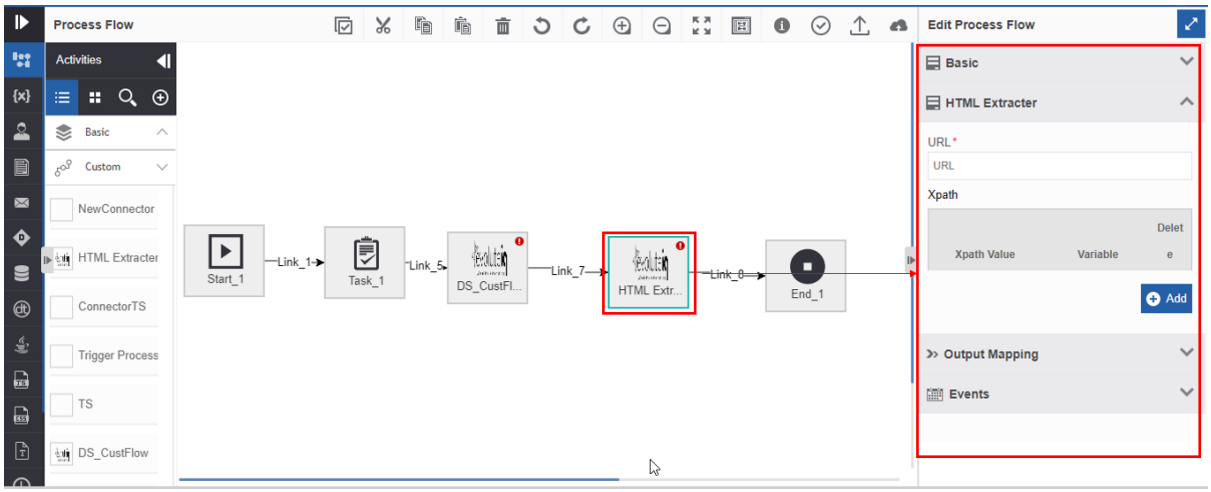


Figure 2.4: Custom activity in process flow and the configuration detail in the Info Actions

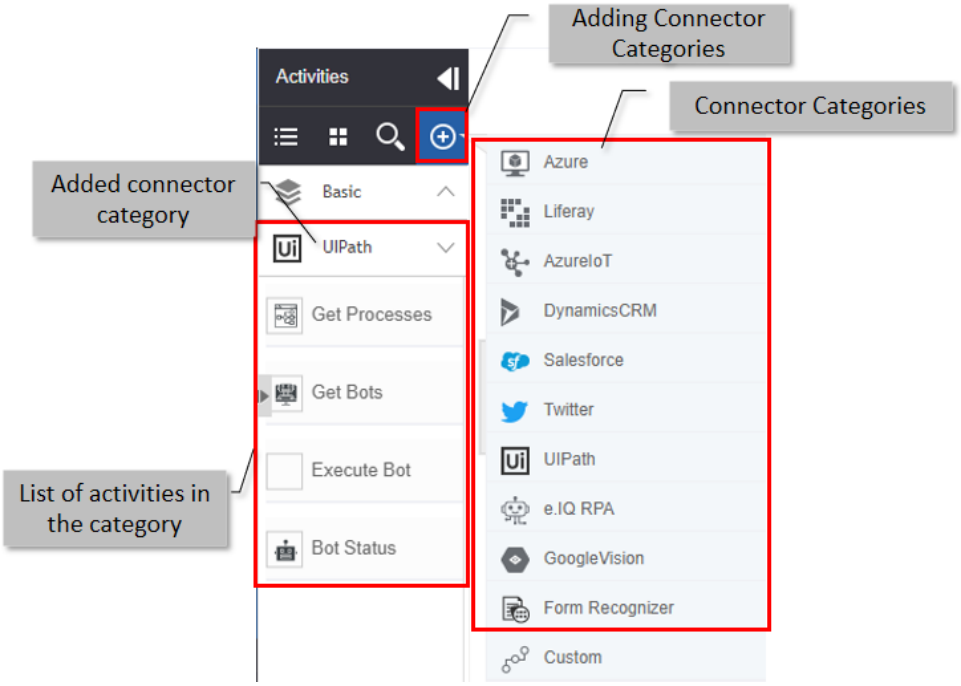
3. Click the activity. Configuration details are displayed on the Info Actions panel.
4. Click **Basic** accordion.
5. Provide **Name** and **Description** for the activity.

6. Click **{custom_configuration}** accordion (that appears on the Info Actions) and configure the details as needed. The custom configuration accordion and details appear based on JSON file details that you provided in the Manage module.

2.3. Connector-Based Custom Activities

The connector-based custom activities are displayed on the specific Connector category in the Activities panel. The connector-based activities (default) and the custom activities can be accessed only by adding those activities to the Activities panel.

- In the Process Flow Activities Panel > Click “+” > click the connector name. All the activities supported for that connector including the custom activities are displayed below the connector category.



3. Appendix

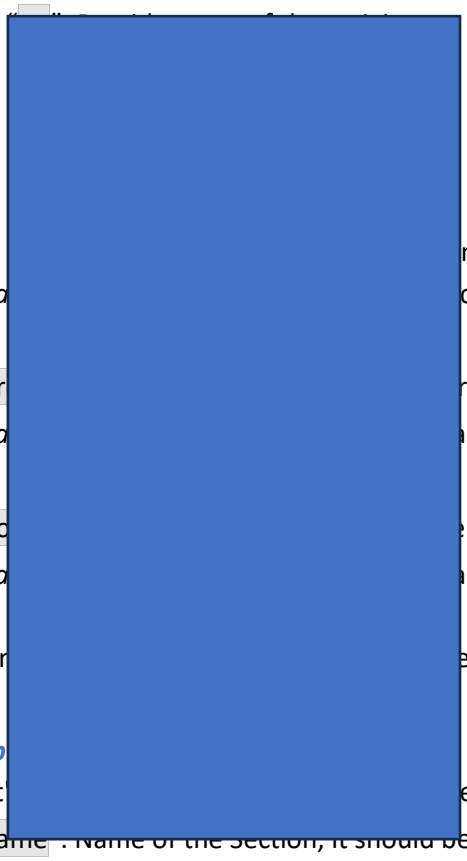
3.1. JSON Code Details

3.1.1. JSON Code Content

Prepare Node JSON as mentioned below.

> *Refer to CustomActivity.json file for sample json format.*

Update the following properties in the JSON Content in the JSON Editor and save.



- ndler class name with full package as mentioned below.

Exa cterHandler"

"pr rty class name with full package as mentioned below.

Exa acterMapping"

"no Property class name with full package as mentioned below.

Exa acterValidator"

By a node configuration: Input Configuration, Output/Error Cor ecated)

Inp

Let e name of the Section.

"name" : Name of the Section, it should be same as a node name.

"tab": Name of the Section without spaces, it will be the accordion name.

"widget": We will define a widget under the widget list.

Each widget will have the following configuration.

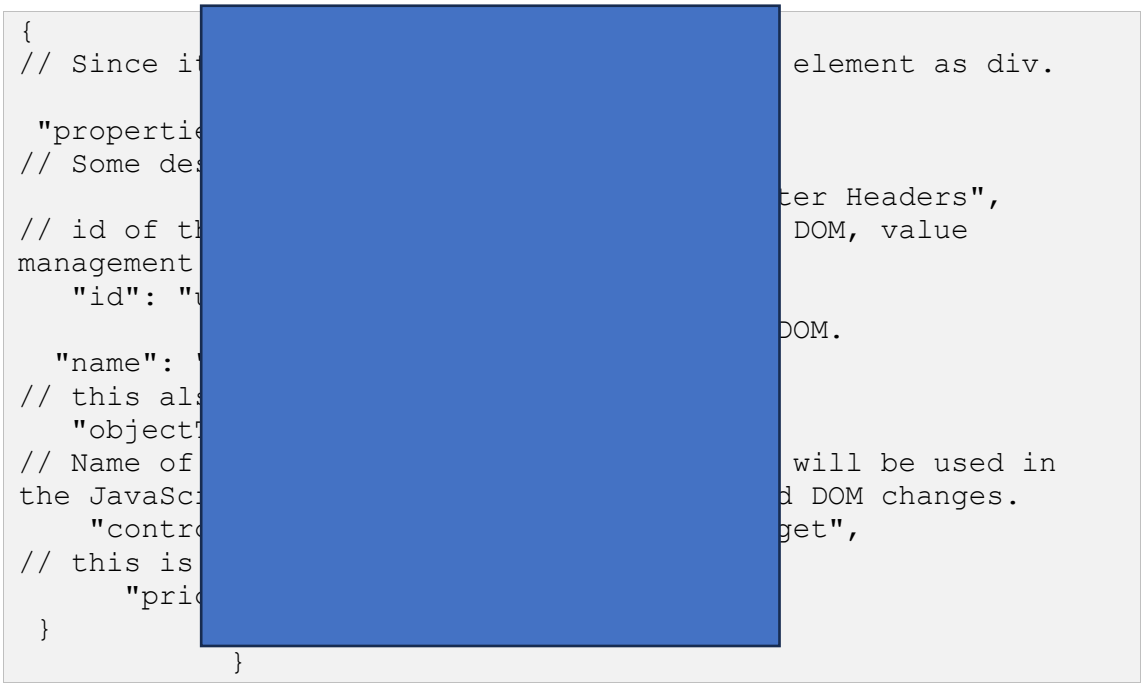
"type": "property-form-custom"

"triggerFunction": "init"

"data": data will have single/multiple objects based on our requirement.

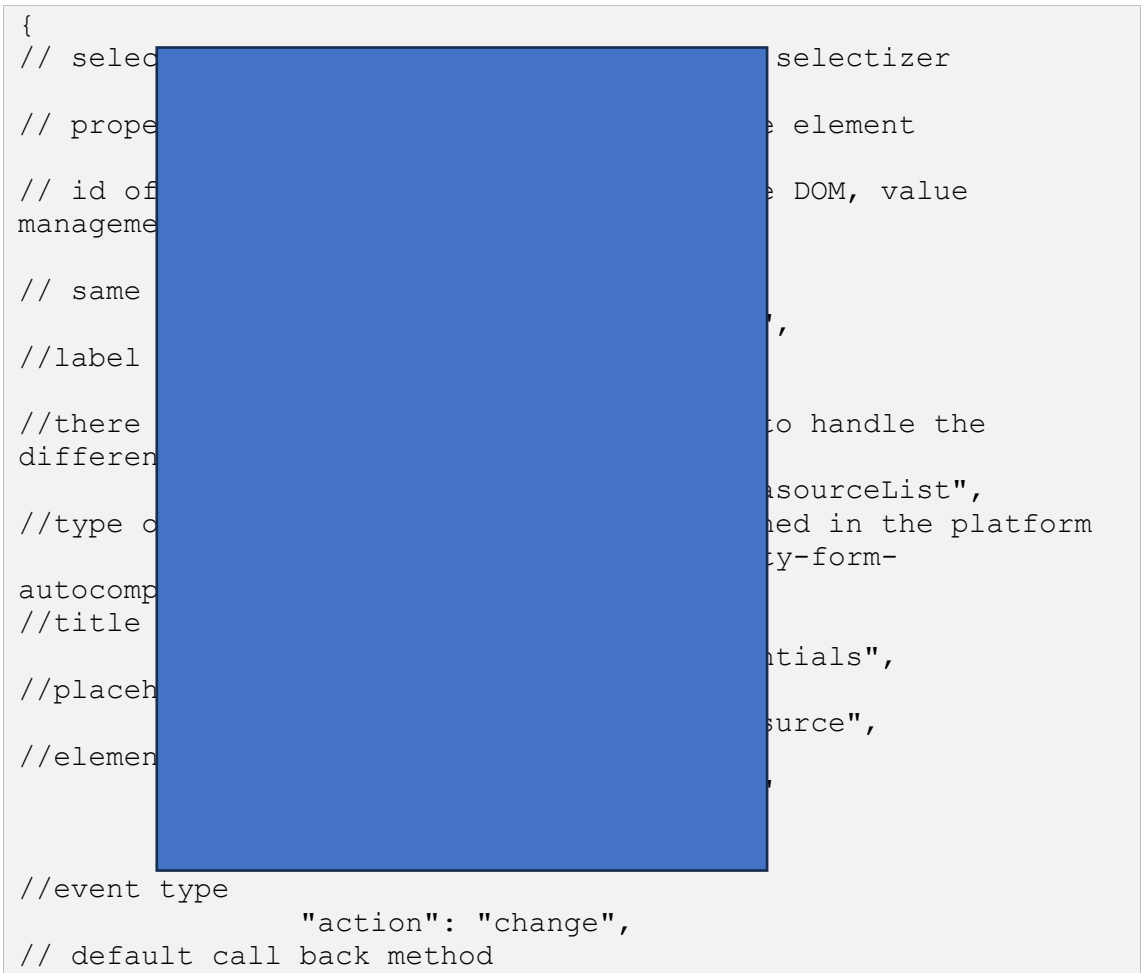
Basically, these are input fields for a Node.

We will define one custom input element under the input section.



Custom Control widget is defined in JavaScript file to handle the DOM changes, events and data management.

Now we will create a Selectize component for listing the configured data sources.



```
                "callBack": "save"
              }
            },
```

Now we will create a normal textbox with droppable variable.

```
{
//input text element
              "e
              "p
//label of the e

//description of

//id of the elem

//class of the e                                    opable class to
handle the varia                                    lidation classes
to validate the

                                                    red,
custom[imagePath
//name of the el

//placeholder of

//type of the in

//titile of the

//default value

//element to be

//variable dropp                                    will be allowed
to drop variable

              },
              "e
//event type

// default call back method
                "callBack": "save"
              }
            }
```

Output/Error Configuration

Use below JSON to create the output and error elements to handle the extracted data or failure cases. This will be commonly used for any custom activity.

```
{
      "name": "Output Mapping",
      "tab": "outputMapping",
      "widget": [
        {
          "type": "property-form-custom",
```

```
            "data": [
              {
                "elementTyp
                "properties
                {
                  "label":                                          e",
                  "descript
                  "id": "ou
                  "name": "
                  "objectTy
mapping",
                  "class":
                  "fieldIde
                  "type": "
                  "placehol
                  "title":
                  "priority
                },
                "event":
                {
                  "action":
                  "callBack
                }
              },
              {
                "elementTyp
                "properties
                {
                  "label":
                  "descript
                  "id": "er
                  "name": "
                  "objectTy
mapping",
                  "class":
                  "fieldIde
                  "type": "
                  "placehol                                         ble",
                  "title":
                  "priority
                },
                "event":
                {
                  "action": "change",
                  "callBack": "save"
                }
              }]
          }]
      },
```

Event Configuration (Deprecated-check before using the code)

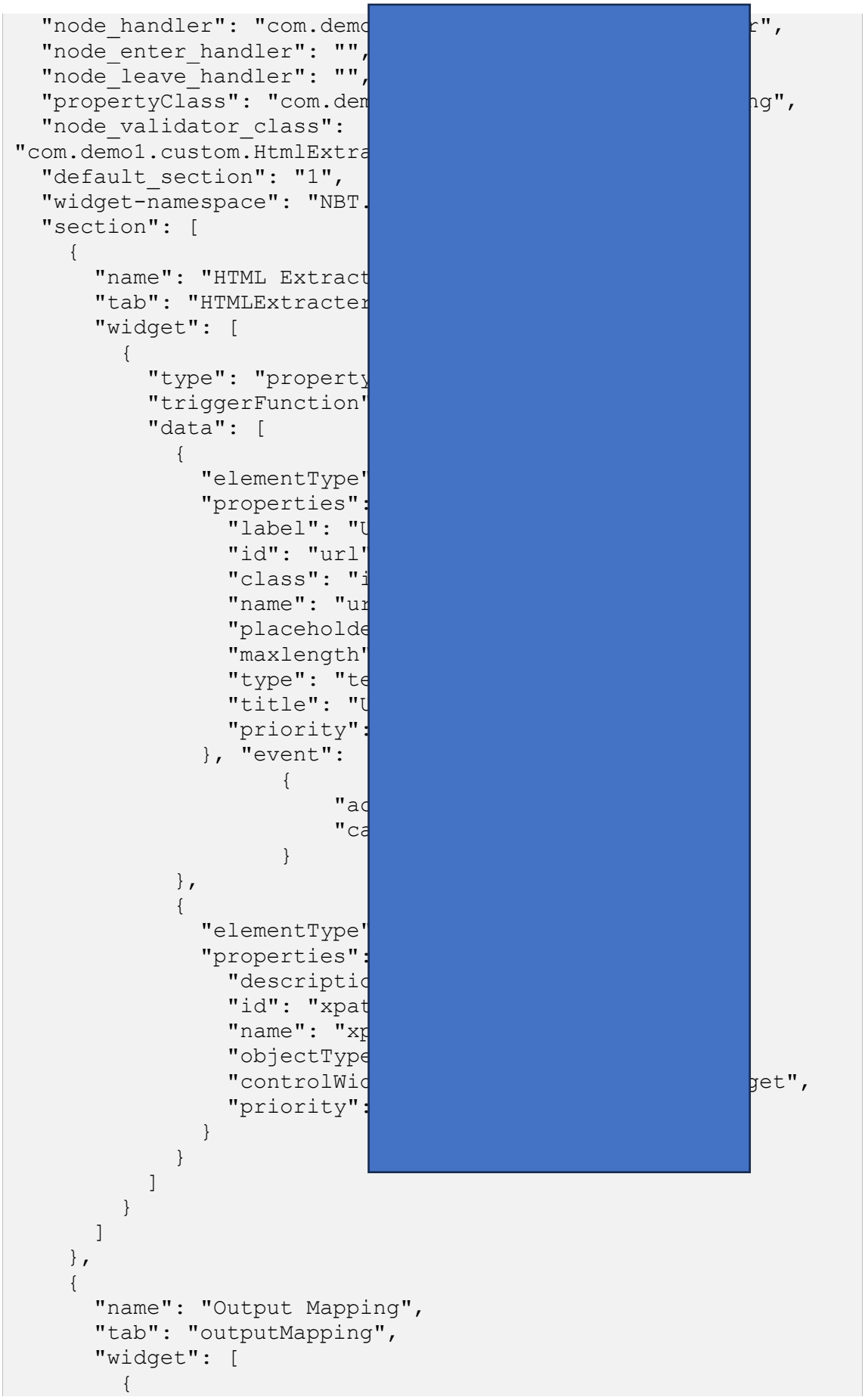
We can use the below JSON to create event configuration.

```
{
      "name": "Events",
      "tab": "property-events",
```

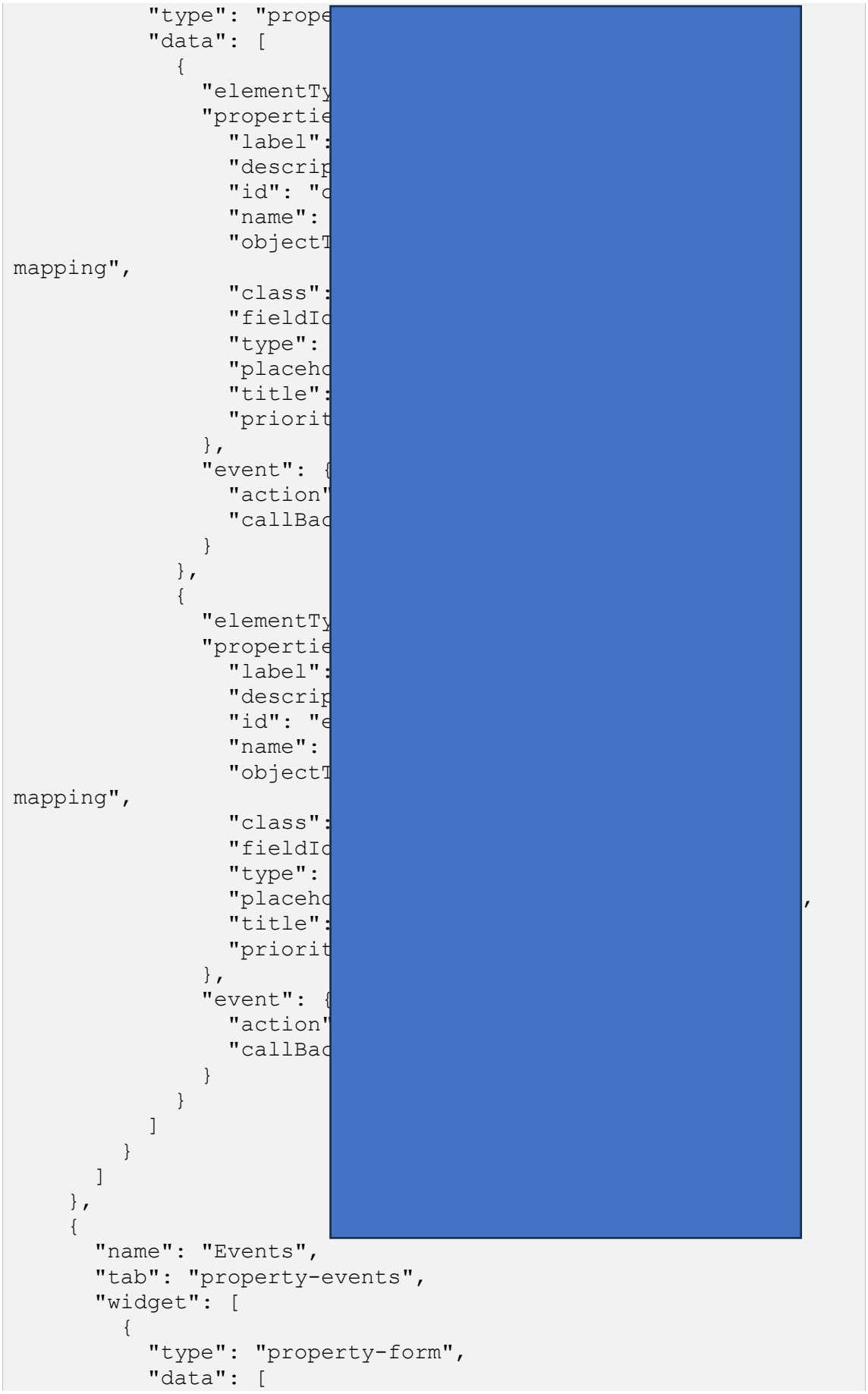
```
      "widget": [
        {
          "type": "property-form",
          "data": [
            {
              "elementT
              "properti
              {
                "id": "
                "name":
                "placeh
                "type":
                "compon
                "compon
                "title"
                "priori
              },
              "event":
              {
                "action": "change",
                "callBack": "save"
              }
            }]
        }]
    }
```

3.1.2. JSON Code Example

```
{
  "nn": "HTML Extracter",
  "nd": "HTML Extracter",
  "loc": "",
  "tp": "svg",
  "it": "true",
  "ot": "true",
  "mit": "true",
  "mot": "true",
  "txt": "HTML Extracter",
  "sts": "",
  "fp": "images/process-designer-node-
icons/nodes/svg/UnitConverter.svg",
  "canvas":
icons/nodes/
  "type": "
  "grp": "Ba
  "group": "
  "ht": "fal
  "nodegroup
  "otc": "1"
  "CKey": "n
  "notValid"
  "grp_icon_
  "grp_icon_                                    r-node-
icons/groups/svg/UnitConverter.png",
```



```
  "node_handler": "com.demo                                          r",
  "node_enter_handler": "",
  "node_leave_handler": "",
  "propertyClass": "com.dem                                          ng",
  "node_validator_class":
"com.demo1.custom.HtmlExtra
  "default_section": "1",
  "widget-namespace": "NBT.
  "section": [
    {
      "name": "HTML Extract
      "tab": "HTMLExtracter
      "widget": [
        {
          "type": "property
          "triggerFunction"
          "data": [
            {
              "elementType"
              "properties":
                "label": "U
                "id": "url"
                "class": "i
                "name": "ur
                "placeholde
                "maxlength"
                "type": "te
                "title": "U
                "priority":
              }, "event":
                    {
                        "ac
                        "ca
                    }
            },
            {
              "elementType"
              "properties":
                "descriptio
                "id": "xpat
                "name": "xp
                "objectType
                "controlWid                                          get",
                "priority":
              }
            }
          ]
        }
      ]
    },
    {
      "name": "Output Mapping",
      "tab": "outputMapping",
      "widget": [
        {
```



```
          "type": "prope
          "data": [
            {
              "elementTy
              "propertie
                "label":
                "descrip
                "id": "c
                "name":
                "objectT
mapping",
                "class":
                "fieldId
                "type":
                "placeho
                "title":
                "priorit
              },
              "event": {
                "action"
                "callBac
              }
            },
            {
              "elementTy
              "propertie
                "label":
                "descrip
                "id": "e
                "name":
                "objectT
mapping",
                "class":
                "fieldId
                "type":
                "placeho                                          ,
                "title":
                "priorit
              },
              "event": {
                "action"
                "callBac
              }
            }
          ]
        }
      ]
    },
    {
      "name": "Events",
      "tab": "property-events",
      "widget": [
        {
          "type": "property-form",
          "data": [
```

```
            {
              "elementTyp
              "properties
                "id": "ev
                "name": "
                "placehol
                "type": "
                "componen
                "componen
                "title":
                "priority
              },
              "event": {
                "action": "change",
                "callBack": "save"
              }
            }
          ]
        }
      ]
    }
  ]
}
```

3.2. JavaScript Code Details

3.2.1. Control Widget JavaScript File

Define the Control widget as mentioned below, this is the default definition of the Control widget.

> *Refer to CustomActivity.js file for sample javascript format.*

```
$.widget("nbt.CustomKeyValuePairWidget", $.nbt.ControlWidget,
{
  options: {
//Define the options required to be reused in the widget
    processVar
},
  _create: fun
//Initialize a
},
render: functi
//define html                                    to create an
element in the
},
_init: functio
//This method                                    s initialised and
available.
}
});
```

3.2.2. Javascript Code Example

```
$.widget("nbt.CustomKeyValuePairWidget", $.nbt.ControlWidget,
{

   options: {
        processVariables:
    },

    _create: function() {

        this.objectType =
this.options.value.proper
        this._readOnly =
(this.options.value.prope                                    ly") ? true
: false;
        this._uniqueId =                                     uniqueId;
        this.constants =
            _labels: {
                "table":
                "variable
                    "key"
                    "valu
                    "head
                    "addB

                },
                "customAt
                    "key"
                    "valu
                    "head                                    ,
                    "addB
                },
                "headers"
                    "key"
                    "valu
                    "head
                    "addB
                },
                "xpathVar
                    "key"
                    "valu
                    "head
                    "addB
                },
                "paramete
                    "key"
                    "valu
                    "head
                    "addB
                },
                "formFields": {
                    "key": "Key",
                    "value": "Value",
```

```
                    "headers": "Form Fields",
                    "addB
                },
                "formFiel
                    "key"
                    "valu
                    "head
                    "addB
                },
                "customQu
                    "key"
                    "valu
                    "head
                    "addB
                },
                "columns"
                    "key"
                    "valu
                    "head
                    "addB
                },
                "inputPar
                    "key"
                    "valu
                    "head
                    "addB
                },
                "deletela
                "name": "
                "dataType
                "primaryK
            },
            _tooltips:{
              "primaryKey'
        }
        };
        if(this.objectTyp
              this._getDat
        }
        if(this.objectTyp
              this._getTab
        }
        this.render();

    },

    /**
     * Loads all the temp

     */

    render: function() {
        this.tpl_wrapper =                                    rvice-
headers-rest form-row" style="display: inline-block;padding:
0px;width: 100% !important;"></div>');
        this.tpl_div = _.template('<div></div>');
```

```
        this.tp                                                    ('<div
style="width: 4                                                    v>');
        this.tp                                                    '<label
style="<%=style                                                    
        this.tp                                                    abel
style="padding:                                                    
class="<%=cssCl                                                    
        this.tp                                                    template('<div
id = "mapper-co                                                    ainer custom-
table-component                                                    
        this.tp                                                    e('<input
class="variable                                                    id %>" name="<%=
name%>" style =                                                    
        this.tp                                                    div
style="margin-b                                                    item"><div
class="delete-g                                                    
        this.tp                                                    template('<div
style="display:                                                    -
box;padding:3px                                                    
        this.tp                                                    lass="button btn
theme-color btn                                                    :right"
id="member-add-                                                    %= buttonLabel
%></div></div>'                                                    

        this.tp                                                    ate('<select
id="<%=id%>" na                                                    >" class =
"select" style                                                     -top:
8px;"></select>                                                    
        this.tp                                                    
style="width:98                                                    %= id %>"
name="<%= name%                                                    
        this.tp                                                    te('<input
type="checkbox"                                                    >"  style =
"width: 98%">')
    },


    _getDataTyp
        this.da
            {

            },
            {
                "value":"String"
            },
            {
                "key":"Date",
                "value":"Timestamp"
            },
            {
                "key":"Decimal",
                "value":"Double"
            },
```

```
                {
                      "key":"Number"
                      "va
                },
                {
                      "ke
                      "va
                }
          ]
    },

    _getTableColumType
          this.dataType
                {
                      "ke
                      "va
                },
                {
                      "ke
                      "va
                },
                {
                      "ke
                      "va
                },
                {
                      "ke
                      "va
                },
                {
                      "ke
                      "va
                },
                {
                      "ke
                      "va
                }
          ]
    },

    _init: function()
          var wrapper =

          var headersLab                                    l({
              text:
this.constants._labels
              style: 'co
          }));

          wrapper.append

          // Create attr
          $(this.element).append(wrapper);

          this._layout();
```

```
        this._setValue();
    },

    /**
     * Creates Layout an
     *
     */

    _layout: function()
        var headerAttrib
$(this.tpl_merger_attrib
        var header = thi
        if(this.objectTy
        header = this._c
        }
        headerAttributes

        if(this.objectTy
        var entityRow =
this._createCustomQueryE
        headerAttributes
        }

        if(this.objectTy
        var entityRow =                                    rue);
        headerAttributes
        }

        var addRowContai
        if(this.objectTy                                   eadOnly
|| this._uniqueId == "mo                                   nly){
         $(this.element)                                   r);
        } else {

$(this.element).append(h                                   (addRowC
ontainer);
        }


    },

    /**
     * Creates add more
     *
     */
    createAddButton: fun
        var addContainer
        addContainer.add                                   );
        var addButton =
        "buttonLabel" :
(this.constants._labels[                                   l)?this.
constants._labels[this.c                                   "
        }));
        var clearFix = $(this.tpl_div());
```

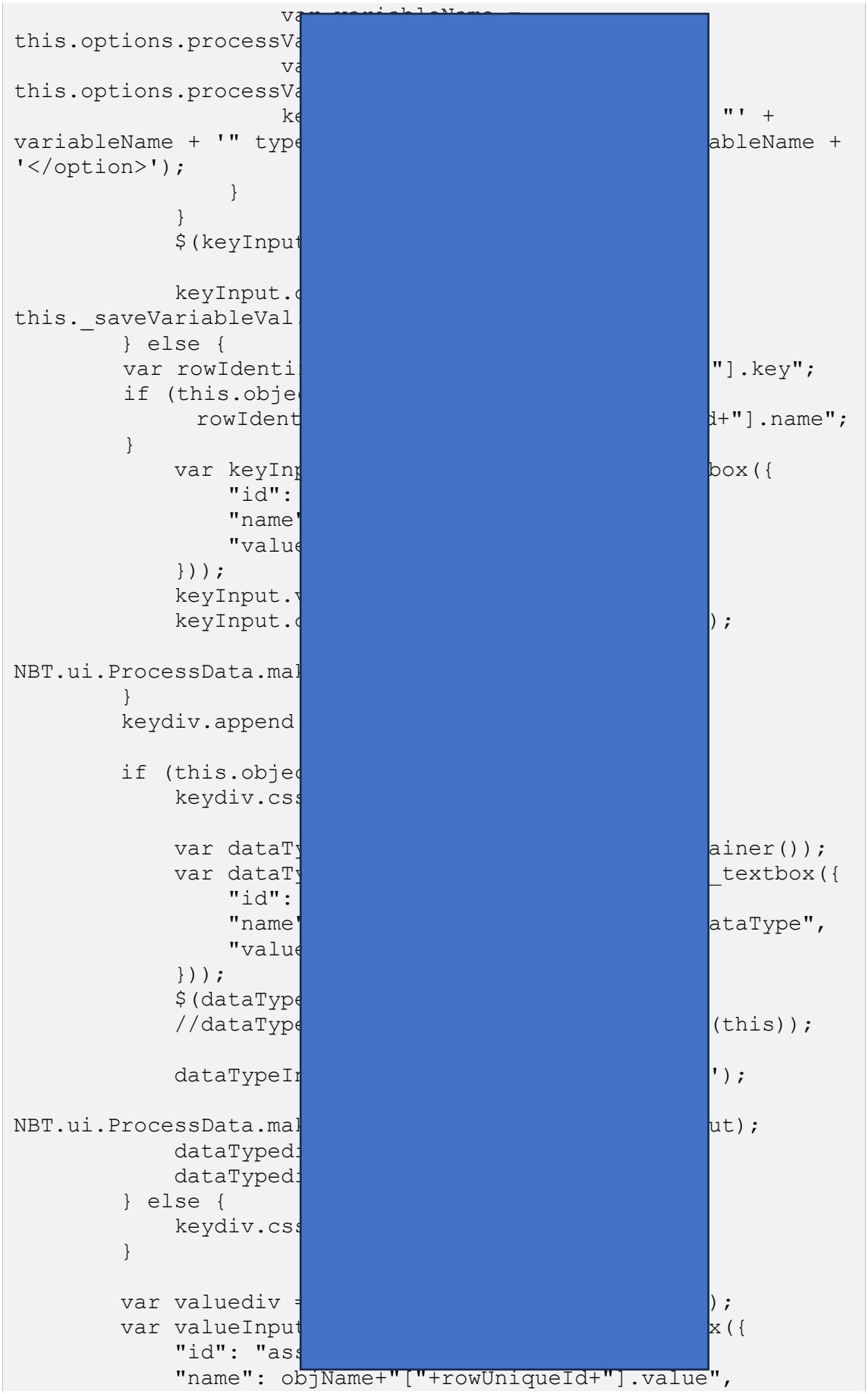
```
        clearFix.addCl
        addContainer.a                                          ix);

        addButton.on('                                          nd(this));

        return addCont
    },

    /**
     * Creates and Add
     *
     * @param {object}
     *            ever
     */

    addItemHandler: fu
        var headerMapp                                          iqueId +
"-container-" + this.c                                          r");
        //this.options
        if (this.optio                                          ables &&
this.options.processVa
        this._getProce
        }
        var row;
        if(this.object
        row = this._ci                                          ;
        } else if(this
        row = this._ci
        } else {
         row = this._
        }

        headerMapperCo
    },

    /**
     * Save the variab
     *
     * @param {object}
     *            ever
     */
    _saveVariableVal:
        if (event && e
            var datat                                           .entity-
mapper-row').find('inp

datatype.val($(event.t                                          'type'));
        }
        this.save();
    },

    /**
     * Get deployed process's variables
     *
     */
```

```
    _getProcessVariables: function() {
        if (this.obje
            var workf
processid").attr("valu
            workflowI                                    kflowId) :
0;

            var appId                                    alue");
            appId = (a
            var _proxy
            var data =
                appli
                proce
            };

            if (appId

_proxy.doRemoteServic                                    .variable
+ '/findByNameAndScope
function(response) {
                    i

$.grep(response.data,
                                                         riable ==
false;


NBT.util.GenericUtil.v                                   Variables,
false, true, true, tru

                                                         s =
processVariables;
                    }
                }.bind
            }
        }
    },

    /**
     * Creates the hea                                   o the
container element
     *
     */

    _createHeaderRow:
        var headingRow
        headingRow.add
        headingRow.css

        var keydiv = $                                   iner());
        var keyConstan
this.constants._labels
        var valueConst
this.constants._labels
        var keyLabel = $(this.tpl_header_label({
```

```
                "text":k
              "cssClass
          }));
          keydiv.append

          if (this.obje
              keydiv.cs
              var datat
$(this.tpl_inline_hea
              var datat                                        bel({
                    "t
                  "css
                }));
              datatyped
              datatyped
          } else {
              keydiv.cs
          }

          var valuediv =                                       tainer());
          var valueLabe
                "text":v
              "cssClass
          }));
          valuediv.appe
          if (this.obje
              valuediv.
          } else {
              valuediv.
          }

          var actiondiv
$(this.tpl_inline_hea
          var actionLab
                "text":t                                       bel,
              "cssClass
          }));
          actiondiv.app
          actiondiv.css

          if (this.obje
          if(this._read
                keydiv.
                valuedi
                datatyp

headingRow.append(key                                          (valuediv)
;
          } else {

headingRow.append(keydiv).append(datatypediv).append(valuediv)
.append(actiondiv);
          }
```

```
        } else if(thi                                                    &
this._readOnly || thi                                                    s" &&
this._readOnly ){
         keydiv.css("
         valuediv.css
         headingRow.a
        } else {

headingRow.append(key                                                    tiondiv);
        }
        return headin
    },

    _createColumnsHea
         var headingR
         headingRow.a
         headingRow.c

         var keyConst
this.constants._label
         var valueCon
this.constants._label

         var primaryK
$(this.tpl_inline_hea
         var primaryK                                                    bel({
              "text":t                                                   ey,
             "cssClass
         }));
         primaryKeydi
         primaryKeydi

         var keydiv =                                                    tainer());
         var keyLabel
              "text":k
             "cssClass
         }));
         keydiv.appen

         keydiv.css("

         var valuediv
$(this.tpl_inline_hea
         var valueLab
              "text":v
             "cssClass
         }));
         valuediv.css
         valuediv.app

         var actiondi
$(this.tpl_inline_hea
         var actionLa
              "text":t                                                   bel,
             "cssClass": ""
```

```
        }));
        actiondiv.ap
        actiondiv.cs
        if(this._rea
        keydiv.css(
        keydiv.css(

headingRow.append(pri                                    end(valuedi
v);
        } else {

headingRow.append(pri                                    end(valuedi
v).append(actiondiv);
        }

        return headi
    },

    /**
     * Creates a new                                     to the
table
     *
     * @param {object
     *            ent
     */

    _createEntityRow

        var rowUniqu
        var objName

        var key = (en                                    ey :
((entity && entity.na
        var dataType                                     ?
entity.dataType : "";
        var value =                                      ty.value :
"";

        var entityRow
        entityRow.add                                    table-
row");

        var keydiv =                                     ;
        if (this.obje
            var keyIn                                    ectBox({
                "id":
                "name                                    name",
                "valu
            }));

            keyInput                                     , ' '));
            if (this.options.processVariables.length > 0) {
                for (var j = 0; j <
this.options.processVariables.length; j++) {
```

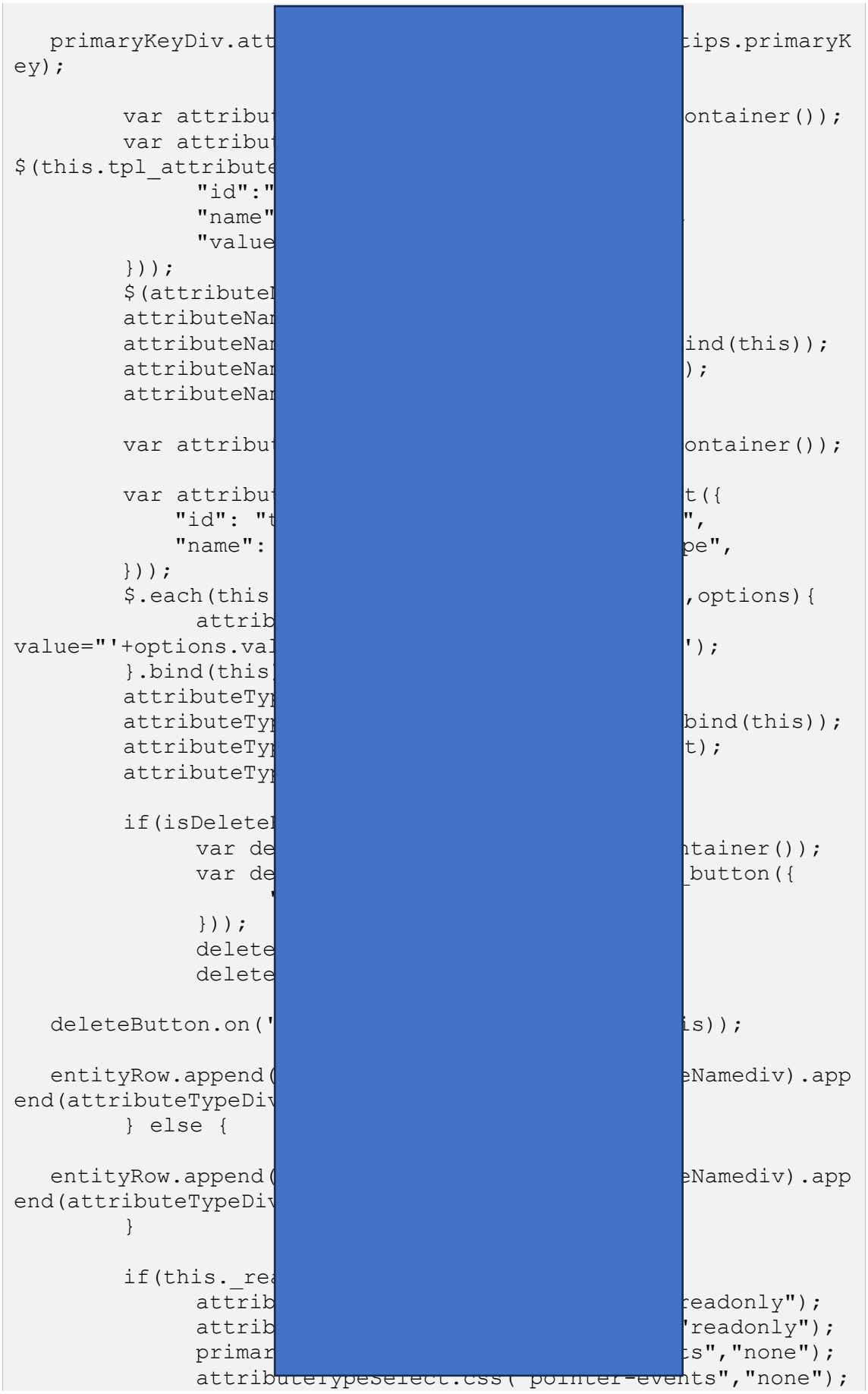


```
                    va
this.options.processVa
                    va
this.options.processVa
                    ke                                          "' +
variableName + '" type                                      ableName +
'</option>');
                }
            }
            $(keyInput

            keyInput.c
this._saveVariableVal
        } else {
        var rowIdenti                                           "].key";
        if (this.objec
              rowIdent                                          d+"].name";
        }
            var keyInp                                          box({
                "id":
                "name"
                "value
            }));
            keyInput.v
            keyInput.c                                          );

NBT.ui.ProcessData.mak
        }
        keydiv.append

        if (this.objec
            keydiv.css

            var dataTy                                          ainer());
            var dataTy                                          _textbox({
                "id":
                "name"                                          ataType",
                "value
            }));
            $(dataType
            //dataType                                          (this));

            dataTypeIn                                          ');

NBT.ui.ProcessData.mak                                          ut);
            dataTypedi
            dataTypedi
        } else {
            keydiv.css
        }

        var valuediv =                                          );
        var valueInput                                          x({
            "id": "ass
            "name": objName+"["+rowUniqueId+"].value",
```

```
            "value": value
        }));
        valueInput.
        valueInput.                                          );
        NBT.ui.Proc                                         alueInput);
        valuediv.ap
        if (this.ob
            valuedi
        } else {
            valuedi
        }

        if (this._u                                         &
this._readOnly) {
            keyInpu
            valueIn                                         );
            if (thi
                key                                        e");
                key
              value
              dataTy

entityRow.append(ke                                        d(valuediv);
            } else
              keydiv
              value
                ent                                        luediv);
            }
        } else {
            var del                                        ainer());
            var del                                        utton({
                "id
            }));
            deleted
            deleted
            deleteB
this._deleteRow.bin
            if (thi

entityRow.append(ke                                        d(valuediv).
append(deletediv);
            } else

entityRow.append(ke                                        eletediv);
            }
        }

        return enti
    },

    /**
     * Creates a new row for the entity and append to the
table
     *
     * @param {object}
```

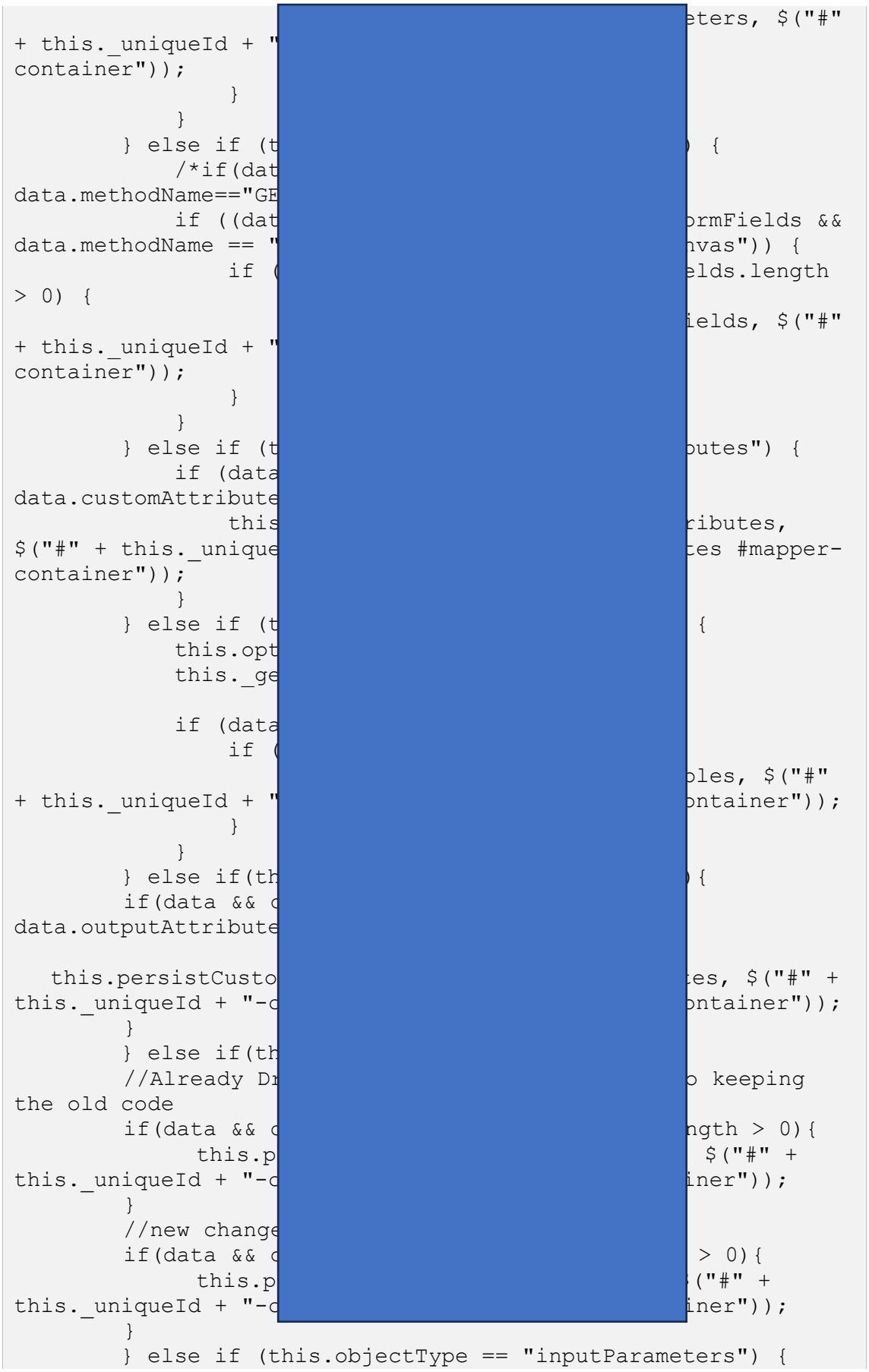
```
     *            entity - Variable object
     */

    _createCustomQueryEntityRow : function(isDeleteButton,
entity){

        var rowUnique
        var name = (e                                    me :"";
        var type = (e                                    pe :"";

        var entityRow
        entityRow.add                                    able-
row");

        var attribute                                    tainer());
        var attribute
$(this.tpl_attribute_t
             "id":"ou                                     .name",
             "name":"                                    "].name",
             "value":
        }));
        $(attributeNa
        attributeName
        attributeName                                    d(this));
        attributeName
        attributeName

        var attribute                                    tainer());

        var attribute                                    {
            "id": "out                                   .type",
            "name": "                                    "].type",
        }));
        $.each(this.da                                   ptions){
             attribut
value="'+options.value                                   ;
        }.bind(this))
        attributeType
        attributeType                                    nd(this));
        attributeTypel                                   ;
        attributeTypel

        if(isDeleteBut
             var dele                                    ainer());
             var dele                                    utton({
                  "ic
             }));
             deletedi
             deletedi

   deleteButton.on('cl                                   ));

   entityRow.append(at                                   teTypeDiv).
append(deletediv);
        } else {
```

```
   entityRow.append(attributeNamediv).append(attributeTypeDiv);
        }

        if(this._readO
             attribute                                    donly");
             attribute                                    adonly");
             attribute                                    ","none");
             attribute
             attribute
        }

        return entityR
    },

    /**
     * Creates a new r                                     the
table
     *
     * @param {object}
     *            enti
     */

    _createColumnsEnti                                    ,
entity){

        var rowUniqueI

        var isPrimary                                     ? true :
false;
        var name = (en                                    e :"";
        //The Key chan
        //To avoid bre                                    les
checking for both type
        var type = (en
entity.dataType : ( ent                                   pe : "";

        var entityRow
        entityRow.addC                                    ble-
row");

        var primaryKey                                    er());
        var primaryKey
$(this.tpl_attribute_c
             "id":"att
             "name":                                      imary"
        }));
        primaryKeyChec                                    (this));
        primaryKeyDiv.
        if(isPrimary){
             primaryKe                                    ;
        }
        primaryKeyDiv.css("width","12%");
```



```
   primaryKeyDiv.att                                          tips.primaryK
ey);

        var attribu                                           ontainer());
        var attribu
$(this.tpl_attribute
             "id":"
             "name"
             "value
        }));
        $(attributeN
        attributeNa
        attributeNa                                           ind(this));
        attributeNa                                           );
        attributeNa

        var attribu                                           ontainer());

        var attribu                                           t({
            "id": "t                                          ",
            "name":                                           pe",
        }));
        $.each(this                                           ,options){
             attrib
value="'+options.val                                          ');
        }.bind(this)
        attributeTyp
        attributeTyp                                          bind(this));
        attributeTyp                                          t);
        attributeTyp

        if(isDeleteI
             var de                                           ntainer());
             var de                                           _button({
                 '
             }));
             delete
             delete

   deleteButton.on('                                          is));

   entityRow.append(                                          eNamediv).app
end(attributeTypeDiv
        } else {

   entityRow.append(                                          eNamediv).app
end(attributeTypeDiv
        }

        if(this._rea
             attrib                                           readonly");
             attrib                                           'readonly");
             primar                                           ts","none");
             attributeTypeSelect.css("pointer-events","none");
```

```
                attribu
                attribu
            }

            return entit
        },

        /**
         * Deletes the s
         *
         * @param {objec
         *             ev
         */

        _deleteRow: func
            if (event &&
                var row
                                                    ity-mapper-
row');
                row.remo
                this.sav
            }
        },

        /**
         * Set values ba
         *
         */

        _setValue: funct
            var data = t
            if (this.obj
                if (data
                                                    REST" &&
data.headers) {
                    if (
                                                    ngth > 0) {
                                                    s, $("#" +
this._uniqueId + "-c
                                                    ner"));
                    }
                }
                if(this.
                                                    ty ==
"upload-tableData"){
                    if (da
                                                    gth > 0) {
                                                    rs, $("#" +
this._uniqueId + "-c
                                                    ner"));
                    }
                }
            } else if (t
                                                     {
                /*if(dat
data.methodName=="GE
                if ((dat
                                                    rameters &&
data.methodName == "
                                                    as") ||
(data && data.method
                                                    ._uniqueId
== "mobileflowcanvas
                    if (data.parameters && data.parameters.length
> 0) {
```



```
                                                                    eters, $("#"
+ this._uniqueId + "
container"));
                    }
                }
            } else if (t                                            ) {
                /*if(dat
data.methodName=="GE
                if ((dat                                            ormFields &&
data.methodName == "                                                nvas")) {
                    if (                                            elds.length
> 0) {
                                                                    ields, $("#"
+ this._uniqueId + "
container"));
                    }
                }
            } else if (t                                            butes") {
                if (data
data.customAttribute
                    this                                            ributes,
$("#" + this._unique                                                tes #mapper-
container"));
                }
            } else if (t                                             {
                this.opt
                this._ge

                if (data
                    if (
                                                                    bles, $("#"
+ this._uniqueId + "                                                ontainer"));
                    }
                }
            } else if(th                                            ){
            if(data && d
data.outputAttribute

   this.persistCusto                                                tes, $("#" +
this._uniqueId + "-c                                                ontainer"));
            }
            } else if(th
            //Already Dr                                            o keeping
the old code
            if(data && d                                            ngth > 0){
                 this.p                                              $("#" +
this._uniqueId + "-c                                                iner"));
            }
            //new change
            if(data && d                                             > 0){
                 this.p                                             ("#" +
this._uniqueId + "-c                                                iner"));
            }
            } else if (this.objectType == "inputParameters") {
```

```
            if (da
data.inputParamete
                th                                  rameters,
$("#" + this._uniq                                  ters #mapper-
container"));
            }
        }else if (                                   {
            if (da                                  ength > 0) {
                th                                  r,$("#" +
this._uniqueId + "                                  ntainer"));
            }
        }
    },


    /**
     * Persist the                                   rmat while
persisting the nod
     *
     * @param {arr
     *                                              bles
     */

    _persistValues                                  ent) {
        $.each(dat
            var en                                  w(entity);
            contai
        }.bind(thi
    },


    /**
     * Persist the                                   rmat while
persisting the nod
     *
     * @param {arr
     *                                              bles
     */
    persistCustomQ                                  s,
containerElement){
        containerE                                  w").remove();
        $.each(att                                  {
             var
             if(i

this._createCustom
             }els

this._createCustom
             }
             cont
        }.bind(this))
    },
```

```
    persistColumnValue :
containerElement){
        containerElement                                    move();
        $.each(attribute
            var entityI
this._createColumnsEntit
            containerEl
        }.bind(this))
    }

});
```

3.3. JAVA Code Details

3.3.1. Installation of Platform JAR Dependencies

Install platform dependencies before we setup any project.

1. Navigate to dependency-repo directory under SupportingFiles in command prompt/terminal.
2. Run the following commands to install the dependencies into maven repository.

```
mvn install:inst                                    .1-sp1.jar -
DgroupId=com.qua                                    ava-node -
Dversion=v5.1.1-
```

```
mvn install:inst                                    1.1-sp1.jar -
DgroupId=com.qua                                    mon -
Dversion=v5.1.1-
```

```
mvn install:inst                                    v5.1.1-sp1.jar -
DgroupId=com.qua                                    nector -
Dversion=v5.1.1-
```

```
mvn install:inst                                    ib-v5.1.1-
sp1.jar -DgroupI                                    artifactId=nbt-
designer-lib -Dv                                    jar
```

```
mvn install:inst                                    .1.1-sp1.jar -
DgroupId=com.qua                                    -mdm-lib -
Dversion=v5.1.1-
```

```
mvn install:inst                                    e-v5.1.1-sp1.jar
-DgroupId=com.qu                                    t-mdm-runtime -
Dversion=v5.1.1-
```

```
mvn install:inst                                    ommonlib-v5.1.1-
sp1.jar -DgroupI                                    factId=nbt-
workflow-commonl                                    aging=jar
```

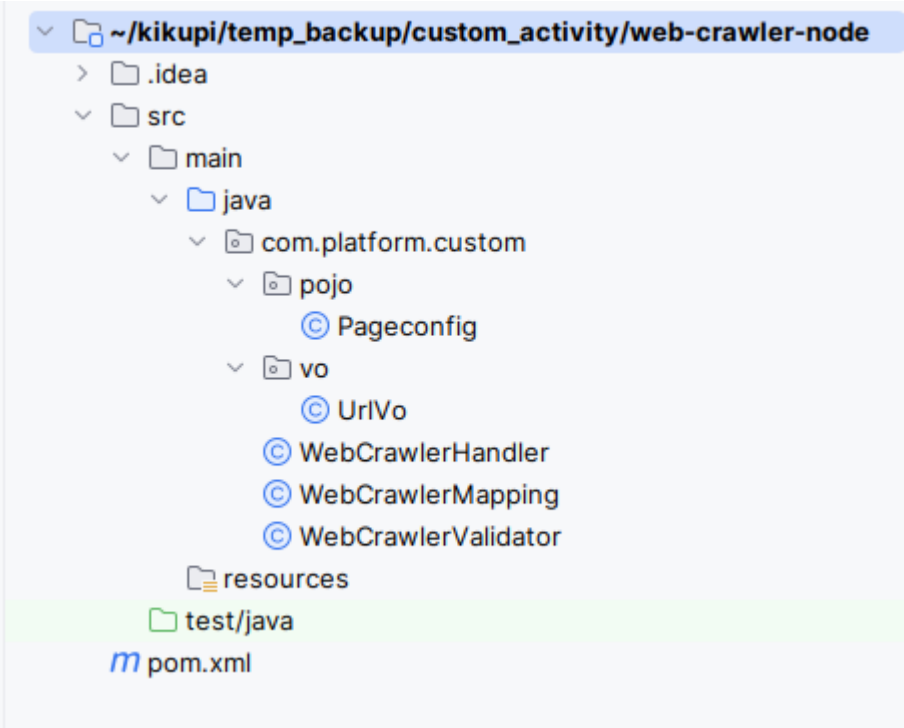
```
mvn install:install-file -Dfile=node-common-v5.1.1-sp1.jar -
DgroupId=com.quadwave.nbt -DartifactId=node-common -
Dversion=v5.1.1-sp1 -Dpackaging=jar
```

For the latest/updated versions of these JARs, contact platform support team.

3.3.2. Supporting Files for Custom Activity

You need to follow a specific structure for organizing the supporting files that are required for the custom activity configuration.

Custom Activity File Organization:



*Refer to provided **JavaJARNode** >**SupportingFiles** folder for file content and details.*

3.3.3. Node JAR Details

The JAR file is mapped in the Node JAR configuration of the Custom Activity.

Node JAR Path

For a custom activity, we will be having a customized java jar, which will have the following classes.

CustomActivityMapping.java

This is the POJO class, we can define all the input and output attributes in the POJO class, which will be mapped to the defined Custom Activity Node. Please find this mapping class in the html-extracter-node project.

CustomActivityValidator.java

This is the validator class, we can define all the mandatory attributes and what type of values and basic validation can be done here in this class.

CustomActivityHandler.java

Whole business logic will be implemented in this class, this class will extend the RuntimeWorkflowNodeHandler.java interface, which is from the Platform end. And we will have to override and provide the implementation for the below methods. Please find this handler class in the html-extracter-node project.

```
public void init(InstanceContext instanceContext);
```

It will set the instanceContext, so that we can have the context of the current instance of the workflow. This context is used to read the variable values at the current moment, and set the variables to the same context, so that it can carry the variable values to the next node in the flow.

```
public HandlerReturn executeNode(NodeContext nodeContext);
```

In this method we will write the business logic, we will read the input data from the Mapping object.

Example:

```
CustomActivityMapping mapping
=(CustomActivityMapping)nodeContext.getBaseProperty();
```

From mapping object, we can read the values and perform any action in this method. Once we are done with our logic, then we can set back the data to the instance.

Below `variableParser` object can be used to read the variable and parse the variable information.

```
VariableParser variableParser = new
VariableParser(instanceContext.getWorkflowInstance());
```

`parseVariableName` method will be use to get the variable name from the mapped object.

```
String outputParam =
variableParser.parseVariableName(mapping.getOutputParam());
```

We can set the value to the instance context as below.

```
instanceContext.setVariableValue(outputParam, "Executed");
```

We can use below method to get the variable value at this moment in the instance level.

```
String variableValue=
instanceContext.getDeductedValue(mapping.getImagePath());
```

Once we get the variable value, we can do implement our logic here.

Once we are done with the business logic, then we can set back the variable values to the instance context, and same way we have to set the Output, and Error data also to the instance context.

Once we are completed with the setting values to the variables, we can call the below code to continue the flow and we have to return the same object as below. Which means the current node handler is completed and control will reach the next node in the workflow.

```
return new HandlerReturnImpl(null, false);
```

We will have Platform predefined service to get the connection related configuration services, like DataSource's, DMS, SMTP, and other connection configuration services will be available to connect and perform some action. Along with these we can have other services like Data Entity Services, MDM Services, And Other factory methods will be available.

Based on our requirement we can use the preferred services.

For example, if we want to persist some data through MDMService, we can follow below steps.

```
//Getting the da
DataEntityServic
MdmServiceRegist

//Getting the da                                    zationId, and
data entity name                                    class.
DataEntity dataE
dataEntityServic                                    getOrganization
Id(), " HTMLPage

// Setting the d                                    ty, so that MDM
service can unde                                    sidered for the
action.
dataEntity.setDa                                    figurationName(
));

//Get MDM Servic                                    tasource, and
name of the data
MdmService mdmSe
MdmFactory.getMd                                    ganizationId(),
dataEntity.getDa
dataEntity.getDa

//this method wi                                    spective table.
mdmService.inser                                    tedBy(),
dataEntity.getNa
JacksonJsonUtil.                                    );
```

3.3.4. Uploading the Node JAR for Custom Activity

This is an additional information for uploading the JAR file in the custom activity.

Node Pallette Icon:

Upload SVG Icon in 20X20 size. Sample icon file is attached. This is used for the listing purpose.

Node Icon:

Upload SVG Icon in 40X40 size. Sample icon file is attached. This will be used in the designer.

Node Accordion Icon:

Upload a PNG Icon in 20X20 size. Sample icon file is attached. This will be used as the accordion icon.

4. References

4.1. Supporting Documents

- Configuration Management
- Application Management